

Analysis of Cervical Cancer and HPV Transmission in India using HPVsim, an agent-based Model for designing Custom Screening-Treatment and Vaccination Strategy tailored to India Specific demography.

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1. Overview

HPVsim addresses WHO's 2020 goals for cervical cancer elimination by introducing an open-source agent-based model in Python. It evaluates strategies, aiding countries in meeting WHO targets with flexibility and user-friendly tools. The WHO's Global Strategy prioritizes vaccination, screening, and treatment, with SAGE endorsing a one-dose HPV vaccine in 2022. Mathematical models, crucial for shaping guidelines, are advanced by HPVsim, which simulates both biological processes and clinical states for a comprehensive approach.

Table 1: Key Features of HPVsim

Feature	Description
Flexibility	Parameterization with country-specific data, sexual networks, HPV genotypes, and immune responses.
User-Friendly	Implemented in Python, with built-in tools, tests, and documentation. Runs efficiently on standard hardware.
Comprehensive Simulation	Captures HPV acquisition, and progression to cervical cancer, and offers high-resolution disease natural history.
Open Source	Transparent model structure, equations, parameter values, and assumptions. Encourages collaboration and reproducibility.

Table 2: HPVsim Use Cases

Use Case	Description
Screening Strategy Comparison	Evaluates health impact and cost-effectiveness of different screening strategies. Illustrates model adaptability.
Vaccination Age and Sexual Behavior Interplay	Analyzes the interaction between vaccination age and assumptions about sexual behavior. Examines vaccination strategy impact on disease outcomes.

Table 3: Conclusion Highlights

Aspect	Summary
Significance	Contributes to global cervical cancer elimination efforts through in-depth analyses.
Future Implications	Lays the foundation for simulating high-resolution biological processes.
Accessibility	Open-source nature ensures transparency, reproducibility, and collaboration.

2. Design and Implementation

HPVsim is a comprehensive simulation model that represents individuals (agents) over time to simulate the transmission of the human papillomavirus (HPV) within sexual networks and the progression of HPV-related conditions such as cervical dysplasia and cervical cancer. The model encompasses several key components and features:

- **Initialization and Maintenance:** The simulation begins with the initialization of a population of agents, which is then dynamically maintained as the simulation progresses over time.
- **Sexual Networks:** Agents interact with each other through sexual networks, reflecting the transmission dynamics of HPV.
- **HPV Genotype Transmission:** The model considers the transmission of multiple co-circulating HPV genotypes.
- **Disease Progression:** Persistent HPV infections are modeled to progress to cervical dysplasia, followed by either spontaneous regression or the development of invasive cervical cancer.
- **Interventions:** Various interventions, such as vaccination, screening, and treatment, are included in the model to assess their impact on HPV-related outcomes.
- **Model Outputs:** The model provides standard outputs, and it offers flexibility for defining and capturing additional outputs to support various research objectives.
- **Additional Features:** The model incorporates additional features, including multiscale modeling, calibration, and the ability to incorporate dynamics related to co-infection with HIV.
- **Software Architecture:** The model is underpinned by a software architecture that supports its functionality and adaptability for research purposes.

In summary, HPVsim is a versatile simulation model designed to simulate the complex dynamics of HPV transmission and related health outcomes, making it a valuable tool for studying and evaluating interventions and understanding the impact of HPV on public health.

2.1 Model Simulations

This section outlines the philosophy and basic setup for running simulations in HPVsim. It highlights the simplicity of creating and running simulations while also emphasizing the importance of understanding the core assumptions and parameters. The table below summarizes the main points:

Key Point	Details
Simulation Ease	Creating and running simulations can be done in as little as four lines of code.
Core Assumptions	The model relies on a set of core assumptions and default inputs.
Parameter Customization	Users can directly modify parameters during simulation creation.

2.2 Demographics and Population

This section discusses how HPVsim handles demographic aspects and population dynamics, including how it models countries and custom settings. Here's a summary table:

Key Point	Details
Demographic Updates	Demographic updates include births, deaths (excluding cervical cancer-related), and migration.
Country Modeling	The model can simulate HPV transmission and disease progression within specific countries.
Automated Workflow	Demographic workflow is automated using data from sources like World Bank and UN WPP.
Custom Settings	Users have the flexibility to customize demographic inputs for specific analyses.

2.3 Sexual Network

This section explains how HPVsim handles the sexual network, including its initialization and updating, as well as options for configuring different types of networks. Here's a summary table:

Key Point	Details
Sexual Network Options	HPVsim offers 'default' and 'random' network options that can be configured by users.
Relationship Types	The 'default' network includes three relationship types: long-term, casual, and one-off.
Parameters for Relationships	Key parameters for relationship types include duration, concurrency, coital frequency, condom usage, participation rates, and age mixing.
Initialization and Updating	The sexual network is initialized at the start of the simulation and updated over time.
Customization	Users can customize sexual network parameters to suit their research needs.

These summaries provide a detailed breakdown of each section's key points, making it easier to understand the key concepts and functionalities of HPVsim.

2.4 HPV Transmission

This formal report outlines the fundamental equations and parameters used to model HPV transmission within HPVsim, providing a comprehensive understanding of the transmission dynamics incorporated into the simulation.

1. Basic Transmission Equation

- HPVsim incorporates a model of HPV transmission within serodiscordant partnerships, involving a per-act probability of transmission (β_g) for each genotype (g).
- The model assumes that an individual can only be infected with a single genotype at a given time. Multiple concurrent infections with different genotypes, however, are allowed.
- HPV infections are assumed to be independent within an individual, regardless of their current infection status. Own- and cross-immune protection following clearance of infection is considered, as discussed in Section "B-cell mediated neutralizing immunity."
- The probability of transmission from an infected person (i) with genotype (g) to a susceptible person (j) within a time step (t) is defined as follows:

$$\lambda_{i,j,g}(t) = (1 - (1 - \beta_g(1 - \text{eff_cond} \times \text{cond}))^{n(t)}) \times (1 - \text{inf_imm}_{j,g}(t))$$

Where:

- β_g represents the per-act probability of transmission of genotype g .

- eff_cond is the assumed efficacy of condoms.

- cond is the probability of condom use within the partnership.

- $n(t)$ is the number of sexual acts within this partnership between time t and $t+dt$.

Note: $n(t)$ can have indices as i,j to capture the interaction between i (the infected individual) and j (susceptible person).

- $\text{inf_imm}_{j,g}(t)$ describes the protective immunity of person j to infection by genotype g .

Note: A person who seroconverts is given a level of immunity against the genotype cleared, they however, may develop a partial protective immunity against another genotype also. This is explained in the next paragraph.

Immunocompromised impacts the likelihood of disease progression, it does not impact likelihood of transmission.

2. B-cell Mediated Neutralizing Immunity

- The probability of HPV transmission is moderated by the immunity of the susceptible partner and the relative transmissibility of the infected partner. B-cell-mediated immunity neutralizes the virus at the point of exposure and can result from either infection or vaccination.
- Within HPVsim, this immunity begins following an individual's clearance of infection with genotype g . Upon clearance, the individual may seroconvert with probabilities that depend on the genotype.
- The individual is then assigned an immunity level ($\text{nab_imm}_{j,g}$) drawn from a beta distribution.
- By default, this immunity level remains constant over time. However, users can configure simulations to include waning immunity with a specified decay function ($\text{nab_imm}_{j,g}(t)$).
- The effective immunity against infection from any genotype h is calculated using a cross-immunity matrix (Table 1) as shown below:

$$\text{inf_imm}_{j,h} = \sum_{i=1}^G \text{nab_imm}_{j,i} * \text{ximm}_{i,h}$$

Where:

- $\text{ximm}_{i,h}$ represents the entry from the i th row and h th column of the cross-immunity matrix ximm .

- This example illustrates how individuals with prior infections benefit from immunity against other genotypes.

- Table S1: Cross immunity matrix

**Note: hrHPV: all other non-16/18 oncogenic HPV genotypes. These values are based on a compilation of evidence from cross-reactivity studies.*

Protection against ↓ after infection by →	HPV16	HPV18	hrHPV
HPV16	1	0.5	0.3
HPV18	0.5	1	0.3
hrHPV	0.3	0.3	1

3. Relative Transmissibility

- HPVsim offers flexibility in allowing an infected person's relative transmissibility to vary based on their HPV infection stage (episomal, transforming, cancerous).
- Default settings treat all stages of infection and dysplasia as equally infectious, except for cancer, which is assumed to be non-infectious.
- Different values of relative transmissibility can be set for infections that are episomal or transforming if desired, facilitating the exploration of time- and health-state-varying viral load and its impact on transmission.

These equations and parameters govern the transmission of HPV within HPVsim, accounting for various factors affecting transmission and immunity.

2.5 Disease Natural History

In HPVsim, persistent episomal HPV infection can lead to cellular transformation, possibly resulting in cervical cancer. Prognoses are predetermined upon infection and can be updated due to interventions or immune changes. The method involves sampling infection duration, assessing severity, and determining clearance or transformation. Irreversible transformation leads to cancer.

This approach minimizes health state changes for efficiency. In males, infection duration is sampled from a log-normal distribution with immediate clearance, excluding persistent infections and cancer outcomes.

1. Episomal HPV Infection and Transformation:

In evaluating the progression and transformation of HPV infection:

Duration of Episomal Infection:

- We model the duration of episomal HPV infection before clearance or transformation as a log-normal distribution, with genotype-specific mean values and distributions.
- These durations are adjusted for each individual based on *cell_immj,g*, which represents the level of immunity an individual possesses against abnormal cell growth from a previously cleared infection.

Infection Severity:

- By default, we assume that the severity of the infection, denoted as s_g , increases over time following a logistic function

$$s_g(t) = \frac{-1}{1 + e^{k_g * c_g}} + \frac{1}{1 + e^{-k_g(t - c_g)}}, \quad k_g \sim N(\mu_g, \sigma_g^2)$$

- This function has genotype-specific parameters, including an inflection point (c) and a growth rate (k), both of which are normally distributed.

Classification into Histologic Categories:

- Infection severity is used to classify women into different histologic categories based on clinical cutoffs. These categories provide insights into the progression of infection.

Probability of Transformation:

- The probability of episomal infection transitioning into cellular integration is a stochastic process influenced by the severity of infection over time.
- It is represented by the equation:

$$\text{prob}(\text{transform} | \text{genotype} = g, \text{dysplasia} = s) = (1 - (1 - \pi_g)^{s_g(t) * 100})$$

Here, π_g represents the genotype-specific probability of a cellular integration event for a single episomal cell. This parameter accounts for variations in oncogenicity among different genotypes.

Irreversibility and Cancer Onset:

- Upon cellular integration, the process is considered irreversible, and if left untreated, it leads to invasive cervical cancer. The time to cancer onset is sampled from a log-normal distribution.
- Default values for the transformation probability and time to cancer are calibrated based on age-specific cancer rates and the distribution of HPV genotypes in cancer cases across various countries.

2. Evaluation of Infection Growth and Transformation:

- Duration of episomal HPV infection follows a log-normal distribution, varying by genotype.
- Infection severity increases over time based on a logistic function.
- The growth function can be customized by users if needed.

3. Probability of Transformation:

- The probability of episomal infection transforming and integrating is a stochastic process.
- It depends on infection severity and genotype aggressiveness.
- Once cellular integration occurs, it leads to irreversible invasive cervical cancer.

4. T-Cell Mediated Immunity:

- T-cell mediated immunity begins after an individual clears an HPV infection and seroconverts.
- Immunity value is drawn from a beta distribution.
- T-cell mediated immunity confers no cross-immune protection to other genotypes.

5. Clearance, Latency, and Reactivation:

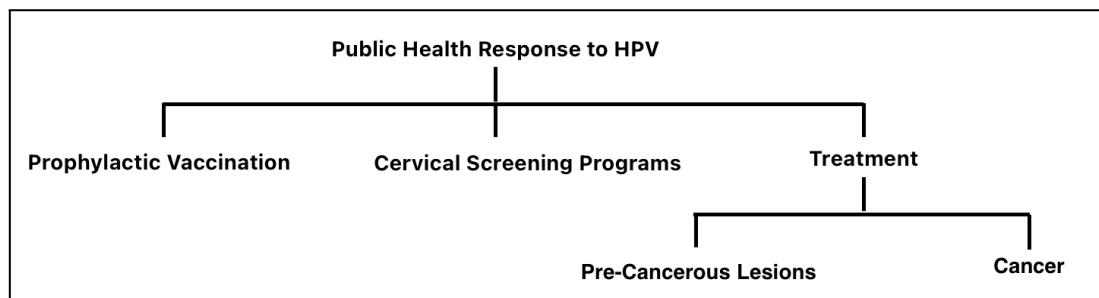
- Clearance represents complete clearance without latent infection.
- Latent infections can be factored in with user-adjustable parameters.
- Individuals who seroconvert develop type-specific B- and T-cell protection.

6. Distribution of Outcomes:

- HPVsim derives an implied distribution of clinical and subclinical outcomes.
- This distribution can be compared to empirical data for validation.
- Outcomes vary by genotype, impacting the incidence of cancer.

2.6. Interventions

Global HPV response integrates vaccination, screening, and treatment. HPVsim enables sophisticated intervention customization and nuanced HIV-related modeling, empowering precise calibration for complex scenarios.



1. Prophylactic Vaccination

Aspect	Details
Efficacy and Dosing	
Vaccine Types	2, 4, or 9 types of HPV with varying cross-protection
Efficacy	High efficacy (97.5%) with single-dose regimen (KEN SHE trial)
Immunity	Single dose confers neutralizing immunity to targeted genotypes
Cross-Immunity	Lower levels of cross-immunity to other genotypes (Table S3)
T-Cell Response	No T-cell response (cell_imm) induced by these vaccines in HPVsim

Duration of Protection	
Long-Lasting Protection	>90% efficacy remains for years: 12 years for quadrivalent, 6 years for nonavalent, and nearly 100% for bivalent
Default Assumption	Immunity from prophylactic vaccination does not wane over time (HPVsim)
Waning Immunity Option	Option to model waning immunity for non-vaccine types
Delivery	
Context-Specific	Delivery details vary by context
Age Group	No default assumptions regarding the targeted age group
Targeted to Girls	Default assumption, but delivery to boys is supported
Routine Vaccination Example	Commenced in 2015, 50% coverage for girls aged 9-10 (routine_vx)
Catch-Up Campaign Example	90% coverage for 15-24-year-old girls/women in 2025 (catchup_vx)
Dosage	Assumed single-dose delivery for efficacy and coverage parameters
Multi-Dose Option	Modeling of multi-dose regimens possible (examples in documentation)

2. Screening and Triage

Aspect	Details
Screening Products	
Purpose	Identify HPV infection or precancerous lesions before invasive cervical cancer
Common Strategies	HPV DNA testing, cytology (Pap smear or liquid-based cytology), visual inspection with acetic acid (VIA)
Outcomes	Possible outcomes defined (positive, negative, inadequate for HPV DNA testing, normal, ASCUS, abnormal for cytology)
Probability	Probability of outcomes given the true underlying disease state (documented in Table S4)
Evolving Technologies	Machine learning-supported visual inspection (AVE), self-collection options
Customization	New screening products can be defined within HPVsim with examples provided in the documentation
Triage Products	
WHO Recommendations	HPV genotyping, colposcopy, VIA, or cytology as triage tests following primary screening
Treatment Pathway	Triage products administer screening tests and assign women to additional triage or treatment based on test results and specified algorithm
Options	Classify women into distinct treatment states, immediate treatment, or repeat screening based on outcomes
Screening and Triage Delivery	

Context-Specific	Delivery platforms vary based on context, including early sexual debut and high HIV burden
Default Assumption	HPVsim assumes screening programs target women aged 30-50 years; users specify screening frequency and eligibility criteria
Example	A routine screening program using VIA started in 2015, with 50% lifetime coverage for women aged 30-50
Eligibility Criteria	Conditions for triage eligibility defined (e.g., returning a positive screening result)
Multi-Step Algorithms	Different components of screening and triage can be linked together to create complex multi-step algorithms

3. Treatment

Aspect	Details
Thermal Ablation	
Procedure	Application of heated probe (100°C - 120°C) to cervix to destroy precancerous lesions
WHO Guidelines	Recommended for histologically confirmed CIN2/3 or screened-positive women
Effectiveness	Highly effective with a 93.8% overall response rate for CIN2+; 91% success rate for lesion clearance according to WHO guidelines
HPV Clearance	Effective in clearing HPV infection with reported rates of 80-85% (default value: 80% clearance in HPVsim)
Excisional Treatments	
Types	Include cold knife conization, LEEP, laser conization; remove cervical samples for examination or treatment
Preference	Preferred in high-income settings for further lab work to detect unobserved cancerous cells
Effectiveness	Assumed 95% effective at removing lesions and 80% effective at clearing active HPV infection in HPVsim
Other Treatments	
Alternative Methods	Some countries use alternative ablation methods like cryotherapy, not included as default in HPVsim but can be defined as new treatment product
Treatment of Invasive Cancer	
Common Options	Surgery, radiotherapy, and chemotherapy for invasive cervical cancer; success rates around 80% in the early stages
Treatment Delivery	
Platform Variability	Treatment delivery varies by context, visits required, and treatment modality; triage strategy must be defined for specific context
Default Options	HPVsim offers 'treat_num' for capped treatments with uptake probability and maximum capacity; 'treat_delay' for delayed treatments

4. Therapeutic Vaccination

Aspect	Details
Therapeutic Vaccine Products	
Current Status	No therapeutic vaccine products on the market, but active research with ongoing clinical trials (59,60)
Default Assumptions	Two-dose regimen targeting HPV16 and 18, 50% efficacy for CIN2/3 regression, 70% for CIN1 clearance, 90% for virological clearance
Immune Memory	Default implementation does not confer neutralizing or T-cell immune memory; adjustable for specific analysis
Therapeutic Vaccine Delivery	
Variability in Delivery	Delivery varies across countries and over time; dependent on vaccine characteristics
WHO Use Cases	High-level use cases: mass-vaccination for entire population; screen-and-vaccinate for HPV-positive women (60)
Default Interventions	'Routine_txvx' for mass vaccination, specifying years, age range, eligibility criteria, and uptake; 'Linked_txvx' for integration with other interventions

5. Combining Interventions into Complex Screen & Treat Algorithms

- Practical interventions harmonize screening, triage, and treatment into multifaceted, sequential algorithms.
- WHO delineates seven algorithms, encompassing both screen-and-treat and screen-triage-and-treat methodologies.
- HPVsim's documentation elucidates the integration of these algorithms via diverse intervention combinations.
- An exemplar in Section Excisional treatments showcases the modelling of the screen-and-vaccinate approach.

6. Custom Interventions

- Custom interventions in HPVsim facilitate dynamic parameter alterations, accommodating nuanced long-term behavioral transitions.
- Users possess the flexibility to delineate their interventions, such as quantifying the effects of broadening antiretroviral treatment accessibility for women living with HIV (WLHIV).
- These interventions empower the approximation of fluctuations in population transmissibility and immunocompromised states, allowing for intricate modeling of epidemiological dynamics.

2.7. Model Outputs

Standard Outputs

Output Type	Description
Flows	Annual counts of new infections, CINs (Cervical Intraepithelial Neoplasia), and cancers.
Stocks	Counts of people in specific disease states at a given point in time.
Demographic Outputs	Includes crude birth and death rates, and population sizes categorized by sex.
Cumulative Aggregates	Totals of individuals who have received interventions like screening, vaccination, or treatment.

Analyzers for Custom Outputs

Analyzer Type	Description
Age Pyramids	Capture age distributions of populations at specific points in time.
Age of Infection	Records the age at which individuals who develop cervical cancer were initially infected with HPV.
Cancer Detection	Calculates rates of cancer detection, providing insights into the effectiveness of cancer screening programs.
Custom Age Groups	Allows for the creation of customized age groups to analyze specific age-related outcomes and statistics.

2.8. Additional Features

1. *Impact of HIV on HPV Pathogenesis*

- HPVsim allows modeling the influence of HIV on HPV acquisition and progression.
- User-provided data includes HIV incidence by age, sex, and time, as well as ART (Antiretroviral Therapy) coverage trends.
- ART "adherence" rates are generated based on specified coverage and life expectancy, reflecting the likelihood of an individual's ART usage.
- HIV infection can modify an individual's risk of HPV acquisition and progression based on ART adherence.
- Prognoses for individuals with prevalent HPV infection are adjusted upon HIV infection, considering relative risk estimates.

2. *Multiscale Modeling*

- HPVsim employs multiscale modeling to address the discrepancy between common HPV infections and rare cervical cancer cases.

- Different agents have various levels of resolution and sampling to balance statistical accuracy and simulation speed.
- Agents are categorized into "Level 0" (highest resolution) and "Level 1" (lower resolution) based on their roles and participation in disease transmission networks.
- This approach optimizes model performance while maintaining statistical precision and can be adjusted as needed.

3. Calibration

- Calibration is the process of configuring the model to match observed data, involving parameter selection that aligns model predictions with real-world outcomes.
- HPVsim offers a built-in calibration function using the Optuna hyperparameter optimization framework.
- Users can specify parameters to calibrate, set bounds based on literature, and provide data files for fitting.
- Calibration aids in adjusting the model to a specific context by fine-tuning parameter values to replicate observed data.

3. Outlook and Possible Modifications

1. Look at India-centric interventions:
 - a. Starting with Screening, Educating and Vaccinating and their combinations
 - b. Optimise the policy as a combination of three interventions with HPVSim as a surrogate model
 - c. Look at the important screening
2. How to include molecular-level information such as population genotype

Results

Use Case 1: Policy Exploration of Screening Algorithms

In this use case, the authors explore the effectiveness and cost-effectiveness of seven screening algorithms recommended by the WHO for cervical cancer prevention.

These algorithms differ in their screening and treatment strategies, and the study focuses on the use of thermal ablation as the treatment method.

The key findings include:

- **Effectiveness:** All seven algorithms substantially reduce cancer incidence compared to no screening, with Algorithm 2 (HPV DNA screen-and-treat) being the most effective due to its high sensitivity.
- **Treatment Rates:** Algorithm 1 (VIA) results in significantly more treatments compared to Algorithm 2, primarily due to the poor specificity of VIA.
- **Triage Benefits:** Triage-based algorithms (3 to 7) reduce overtreatment but may introduce delays and lower treatment coverage.
- **Cost-effectiveness:** Algorithms 3 to 7 are dominated by Algorithms 1 and 2, with Algorithm 1 having a lower incremental cost-effectiveness ratio (ICER) of \$550 per DALY averted compared to Algorithm 2 with an ICER of \$3,500 per DALY averted.

These findings support the use of Algorithm 2 and demonstrate the importance of triage in reducing overtreatment.

Use Case 2: Exploration of Sexual Behavior and HPV Vaccination

In the second use case, the authors investigate the impact of catch-up HPV vaccination campaigns for girls and young women aged 15-24 in three different country archetypes with varying ages of first sexual activity. The key findings include:

- **Context:** The WHO recommends HPV vaccination for girls aged 9, but coverage targets are not always met, necessitating catch-up campaigns.
- **Simulation Setup:** The study uses the HPVsim model and runs simulations from 1950 to 2060, considering demographic data based on Nigeria.

- **Scenarios:** Two scenarios are examined: routine vaccination for girls aged 9-14 starting in 2025 and a supplementary catch-up campaign in 2025 for 50% of girls/young women aged 15-24.

Results: The simulations reveal varying rates of HPV exposure based on the age of first sexual activity and demonstrate the potential health impact of catch-up vaccination for older age groups.

The study highlights the importance of understanding sexual behavior and its impact on the effectiveness of HPV vaccination catch-up campaigns.

Tables and Charts

Table 2: Provides an overview of the screening modalities used in the seven WHO-recommended screening algorithms.

	Primary screen	Triage screen	Triage referral
Algorithm 1	VIA	n/a	n/a
Algorithm 2	HPV DNA test	n/a	n/a
Algorithm 3	Cytology	HPV DNA test	Colposcopy
Algorithm 4	HPV DNA genotype test	VIA for other HR+	n/a
Algorithm 5	HPV DNA test	VIA	n/a
Algorithm 6	HPV DNA test	Colposcopy	n/a
Algorithm 7	HPV DNA test	Cytology	Colposcopy

Table 3: Lists hypothetical costs of various screening tests and treatment products.

Product	Cost (2022 USD)
VIA	\$13
HPV DNA test	\$20
HPV DNA genotype test	\$30
Cytology	\$20
Colposcopy	\$20
Thermal ablation	\$5

Figure 5: Depicts the results of the screening algorithms, including reductions in cancer incidence and the number of ablative treatments over time.

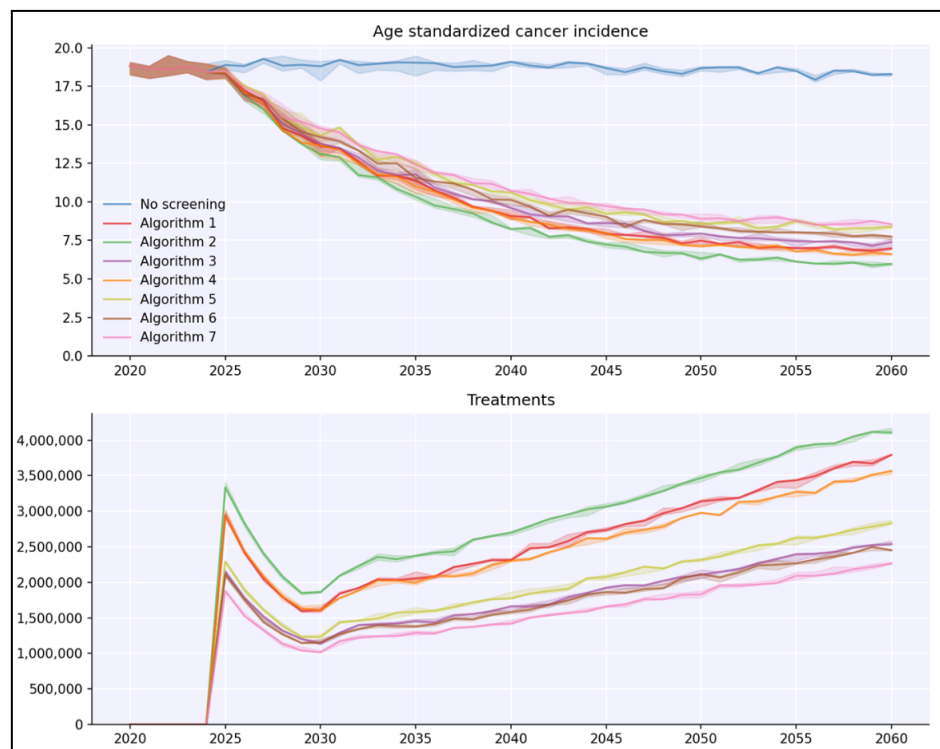


Figure 6: Illustrates the total costs and DALYs averted for each algorithm, showing the incremental cost-effectiveness ratio (ICER) for Algorithm 1 and Algorithm 2.

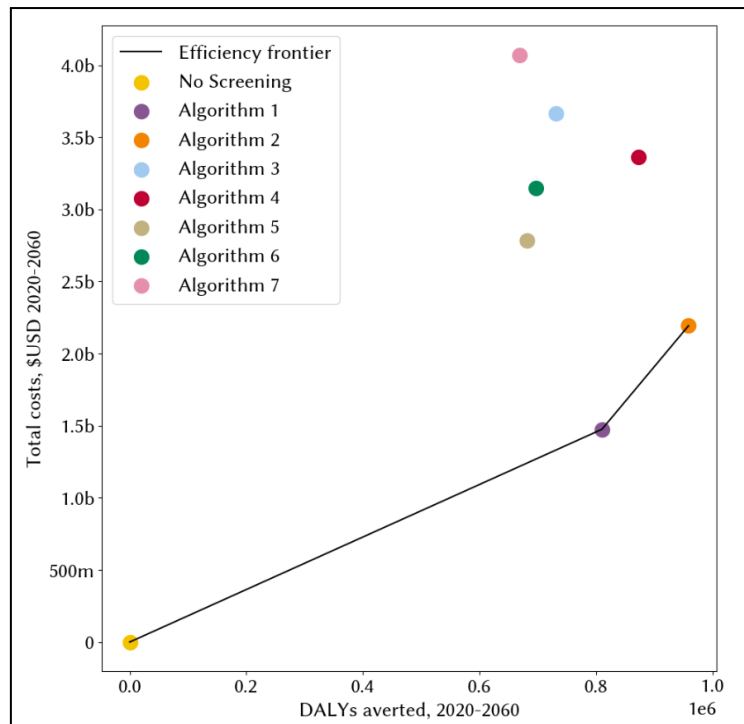
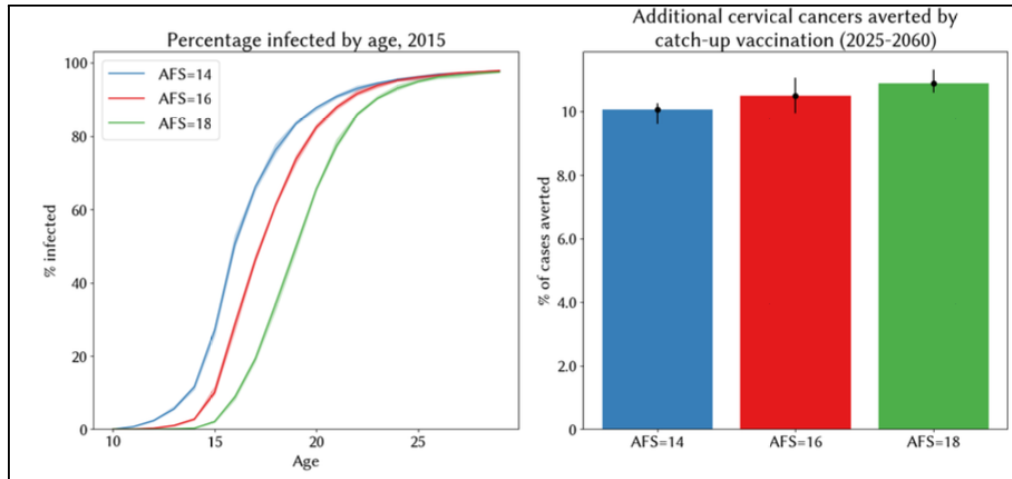


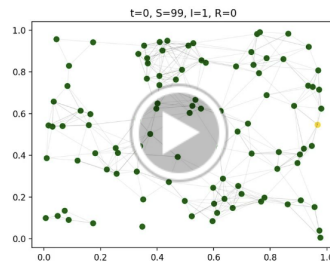
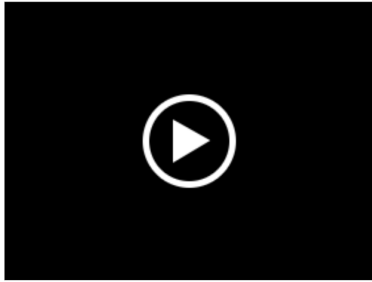
Figure 7: Displays the results of the sexual behavior and HPV vaccination simulations, highlighting the impact on HPV exposure and health outcomes based on age of first sexual activity and catch-up vaccination.



4. Comments

NDMC1.2 - Models for complex diseases_Cliff

Solution: Impact of network



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Ganesh Ramakris...
16:53 Yesterday

SIR kind of behavior. More appropriate for compartmental models.

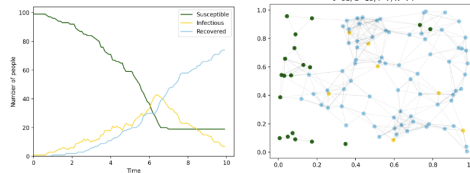


Ganesh Ramakris...
16:53 Yesterday

Bursty and heterogeneous behavior. More appropriate for agent based models.

Exercise: Impact of network

- Consider this network model in Python:
https://github.com/hpvsim/ndmc_training/blob/main/network_model.py
- By modifying and rerunning the script, answer the following questions:
 - What is the impact of changing the "distance" parameter, which controls how long range the connections are? Try values of 100, 1, 0.1, and 0.01.
 - What is the impact of the number of contacts? Try setting n_contacts to 2, 5, 10, and 20.



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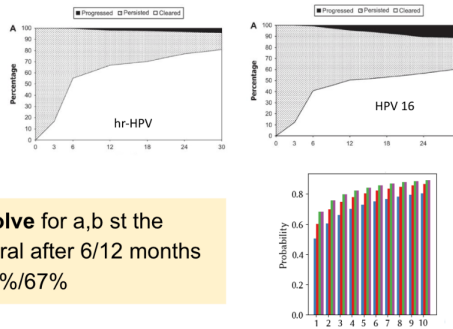
Ganesh Ramakris...
16:47 Yesterday

https://github.com/hpvsim/ndmc_training

NDMC2.1 - Overview of HPVsim_Jamie

Parameterizing duration of infection

- Values from Costa Rica study ([Rodriguez et al 2013](#))
- 800 carcinogenic HPV infections monitored over 30m
 - 55% (52-59%) cleared by 6m
 - 67% (63-70%) cleared by 12m
- Among HPV16 infections:
 - 40% by 6m
 - 50% by 12m



1. Define
 $\text{dur} \sim \log N(a, b)$

2. Solve for a,b st the
integral after 6/12 months
is 55%/67%

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Ganesh Ramakris...
10:04 AM Today

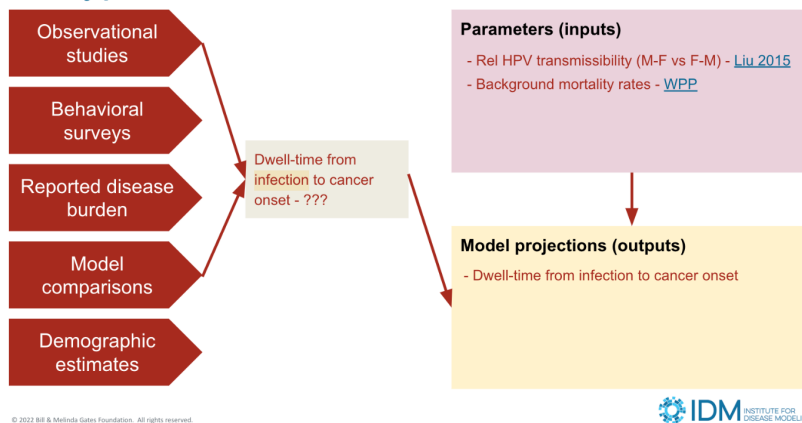
Best received from vaccination drive.

Ganesh Ramakris...
10:05 AM Today

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3705579/>

NDMC2.2 - HPVsim data requirements_Robyn

Data types and uses



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IDM INSTITUTE FOR DISEASE MODELING

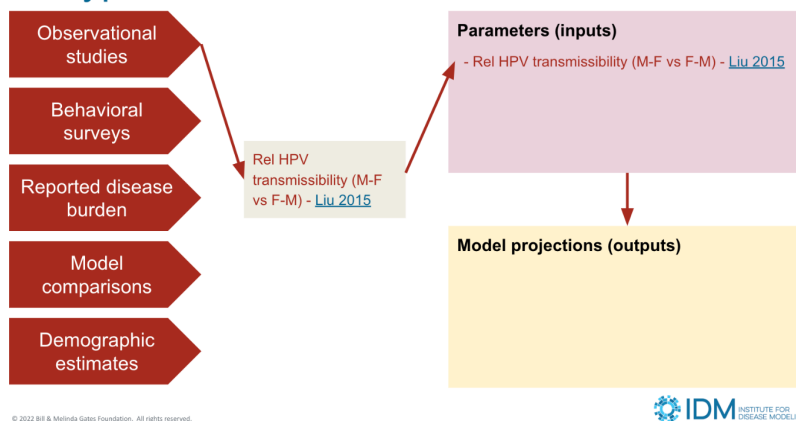
Ganesh Ramakris...
11:15 AM Today

Ideally from registry we should have had this as a "Reported Disease Burden" (infection = high grade lesion). But unfortunately, data in the registry is incomplete.

Ganesh Ramakrishnan
11:18 AM Today

Also for each genotype, the speed and probability changes. No multiplier effect evidence on co-infection based on multiple genotypes.

Data types and uses



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IDM INSTITUTE FOR DISEASE MODELING

Rashmi Tiwari
11:09 AM Today

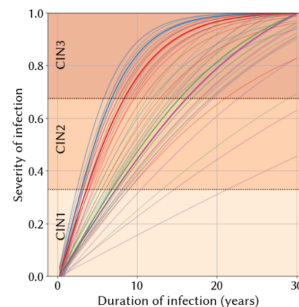
<https://pubmed.ncbi.nlm.nih.gov/26204471/>

Observational data

NDMC2.3 - HPVsim calibration_Robyn

Calibrating individual variability

- Not everyone progresses through stages of cervical disease at the same rate
- HPVsim has a `sev_dist` parameter that captures individual variability
- Adjusting the mean & variance of this distribution can also be helpful for calibration
- Can be thought of as an immuno-compromise parameter



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Ganesh Ramakris...
12:00 PM Today
Immuno compromise parameter To factor in host pathogen interaction?

HPVsim parameters

Basic configuration

- Population parameters
 - Handled by 'location'
- Initialization parameters
 - Handled by dynamics
- Simulation parameters
 - Start & end years

Natural history assumptions

- Immunity
- Control vs clearance

Parameters that need input data

- Network parameters
 - Age of debut, rates of marriage, concurrency

Parameters that need calibrating

- Beta (per-contact transmission)
 - Unmeasurable, uncertain
- Duration of infection by genotype
- Relationship between infection & severity
 - Mean, plus individual variability

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Ganesh Ramakris...
11:54 AM Today
Focus of this session

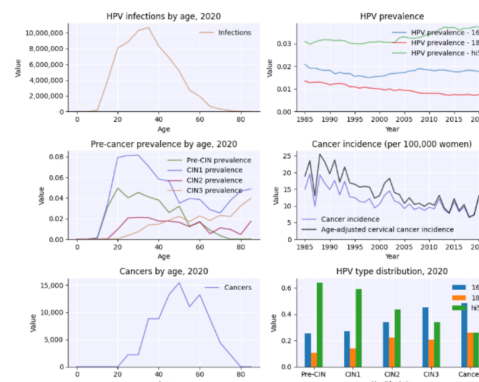
Ganesh Ramakris...
11:59 AM Today
Can we go beyond just HPV genotype to the genetic composition of the population? Immuno compromise parameter To factor in host pathogen interaction?

Customizing local inputs

- With location = 'india', the sim will use India-like demographic parameters

```
import hpvsim as hpv # Import HPVsim
sim = hpv.Sim(location='india') # Make sim
sim.run() # Run the simulation
sim.plot() # Create 6-panel plot
```

- However, the results of this simulation will not necessarily resemble India's HPV and cervical cancer burden - need more local customization.



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Ganesh Ramakris...
11:53 AM Today
Need more local customization beyond setting location='india'

NDMC2.4 - Vaccination in HPVsim_Jamie

Default interventions in HPVsim

Products

Contains the mechanism of action of the product

- Tests – return a test result based upon underlying health state and test positivity matrix
- Vaccines – alter individual immunity attributes
- Treatments – change health state based upon efficacy

Retain/record information on number of products used (useful for CEA analyses)

Delivery

Who, when, how, and what to deliver

- Who: eligibility defined by age, sex, prior outcomes, etc.
- When: start year, interval/frequency, per-year probability
- What: product being delivered
- How: campaign vs. routine (separated for CEA-type analyses)
- Retains information on outcomes i.e., HPV positive



Ganesh Ramakris...
2:35 PM Today

Includes Portfolio Optimization



Ganesh Ramakris...
2:45 PM Today

Portfolio Optimization includes these and possibly more

Defining a custom intervention

```
import hpvsim as hpv
class change_debut(hpv.Intervention):
    def __init__(self, start=None, new_debut=19, *args, **kwargs):
        super().__init__(**kwargs) # NB: This line must be included
        self.start = start
        self.new_debut = new_debut
        return

    def initialize(self, sim):
        super().initialize() # NB: This line must also be included
        return

    def apply(self, sim):
        if sim.t == self.start:
            sim.pars[debut]['f']['par1'] = self.new_debut
            return
```



Ganesh Ramakris...
2:20 PM Today

Sexual education might reflect in concurrency, age debut, casual relationship, especially in a PHASED manner. In the composition, you can spell out the PHASING. In Vaccination and Screening, you need to align.



Ganesh Ramakris...
2:28 PM Today

Another consideration: Screening does not have the HERD mentality benefit that Vaccination has.

However, screening is what will give the immediate benefit and is what will draw attention of the policymaker.

Screening at HPV level might have more multiplier level than Cervical cancer level.

[Show less](#)

NDMC2.5 - Screen & treat in HPVsim_Robyn

Screening interventions - inputs

- product
 - any predefined label from within the [products_dx.csv](#) file
 - an instance of a `hpv.Product`
- prob
 - Probability of a woman being screening (lifetime)
- eligibility
 - who is eligible
- start_year (for routine delivery) or year (for campaign delivery)
- age_range: default 30-50

Parameters

```
primary_screen_prob = 0.2
start_year = 2025
screen_eligible = lambda sim:
    np.isnan(sim.people.date_screened) |
    (sim.t > (sim.people.date_screened + 5 /
sim['dt']))

via_primary = hpv.routine_screening(
    product='via',
    prob=primary_screen_prob,
    eligibility=screen_eligible,
    start_year=start_year,
    label='via primary',
)
```

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Eligible to be screened if you were never screened



Ganesh Ramakris...

3:57 PM Today

OR Eligible to be screened if you were screened more than 5 years back



Ganesh Ramakris...

3:57 PM Today



Default... No symptoms known!
Lesions sitting in cervix but because it is asymptomatic, one would not know unless it turned into cancer.



Ganesh Ramakris...

4:06 PM Today

Targetted screening would largely need to get inputs on the sexual contact network 15-20 years back!



Ganesh Ramakris...

4:14 PM Today

Under hpvsim/data/product_dx.csv

Proceedings Doc Comments

1. Basic Transmission Equation

- HPVsim incorporates a model of HPV transmission within serodiscordant partnerships, involving a per-act probability of transmission (β_g) for each genotype (g).
- The model assumes that an individual can only be infected with a single genotype at a given time. Multiple concurrent infections with different genotypes, however, are allowed.
- HPV infections are assumed to be independent within an individual, regardless of their current infection status. Own- and cross-immune protection following clearance of infection is considered, as discussed in Section "B-cell mediated neutralizing immunity."
- The probability of transmission from an infected person (i) with genotype (g) to a susceptible person (j) within a time step (t) is defined as follows:

$$\lambda_{i,j,g}(t) = (1 - (1 - \beta_g(1 - \text{eff_cond} \times \text{cond}))^{n(t)}) \times (1 - \text{inf_imm}_{j,g}(t))$$

Where:

- β_g represents the per-act probability of transmission of genotype g .
- eff_cond is the assumed efficacy of condoms.
- cond is the probability of condom use within the partnership.
- $n(t)$ is the number of sexual acts within this partnership between time t and $t+dt$.

Note: $n(t)$ can have indices as i,j to capture the interaction between i (the infected individual) and j (susceptible person).

~~$\text{inf_imm}_{j,g}(t)$ describes the protective immunity of person j to infection by genotype g .~~

Note: A person who seroconverts is given a level of immunity against the genotype cleared, they however, may develop a partial protective immunity against another genotype also. This is explained in the next paragraph.

Immunocompromised impacts the likelihood of disease progression, it does not impact likelihood of transmission.

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Immunocompromised impacts the likelihood of disease progression, it does not impact likelihood of transmission.

Selected text | `inf_immj,g(t)` describes the pro...



Ganesh Ramakrishnan

13:00 Yesterday

How is this related to the immuno-compromise level?

See <https://docs.google.com/presentation/d/14V5GM0VqZcBdN69SYrNSFZwaqx16LpGs/edit#slide=id.p17>

Ans: immuno-compromise level does not impact likelihood of transmission but the likelihood of progression within an individual).

But: `inf_immj,g(t)` describes Immunity (from vaccination or from previous infection) from acquiring HPV is what is currently in the code base of HPVSim



Yashi

17:03 Yesterday

Done



Reply or add others with @

But: `inf_immj,g(t)` describes Immunity (from vaccination or from previous infection) from acquiring HPV is what is currently in the code base of HPVSim



Yashi

17:03 Yesterday

Done



Reply or add others with @



Ganesh Ramakrishnan

12:57 Yesterday

$n(t)$ should ideally also have i and k as indices to capture interaction between i and j



Yashi

16:54 Yesterday

Done



Reply or add others with @

it easier to understand the key concepts and functionalities of HPVsim.

2.4 HPV Transmission

This formal report outlines the fundamental equations and parameters used to model HPV transmission within HPVsim, providing a comprehensive understanding of the transmission dynamics incorporated into the simulation.

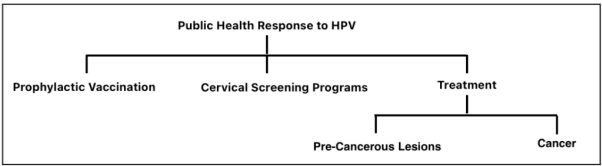
1. Basic Transmission Equation

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2.6. Interventions

Global HPV response integrates vaccination, screening, and treatment. HPVsim enables sophisticated intervention customization and nuanced HIV-related modeling, empowering precise calibration for complex scenarios.



1. Prophylactic Vaccination

Aspect	Details
Efficacy and Dosing	
Vaccine Types	2, 4, or 9 types of HPV with varying cross-protection
Efficacy	High efficacy (97.5%) with single-dose regimen (KEN SHE trial)
Immunity	Single dose confers neutralizing immunity to targeted genotypes
Cross-Immunity	Lower levels of cross-immunity to other genotypes (Table S3)
T-Cell Response	No T-cell response (cell_imm) induced by these vaccines in HPVsim

Selected text | 2.4 HPV Transmission

nirav bhatt
14:55 Yesterday

HPV transmission can change with evolution of the virus. It is not important to introduce HPV evolution and its role in transmission rate and its ability to achieve resistance to the vaccination.

Reply or add others with @

Selected text | Episomal HPV Infection and Transfor...

nirav bhatt
14:52 Yesterday

Episomal HPV infection is but difficult to obtain for a given population. How should we solve this problem? Use NZ distribution for all populations.

Reply or add others with @

Selected text | 2.6. Interventions

Episomal HPV infection is but difficult to obtain for a given population. How should we solve this problem? Use NZ distribution for all populations.

Reply or add others with @

Selected text | 2.6. Interventions

nirav bhatt
14:46 Yesterday

We should investigate a combination of screening, sex education (or behaviour modification), and vaccination. For the screening, the model can be used for identifying the pockets (or) high priority age groups for limiting spread. We can also use the concept of dynamic optimization or open-loop optimal control for finding an optimal combination of different interventions for given budget/resources.

Reply or add others with @

5. Analysis of HPVsim Software

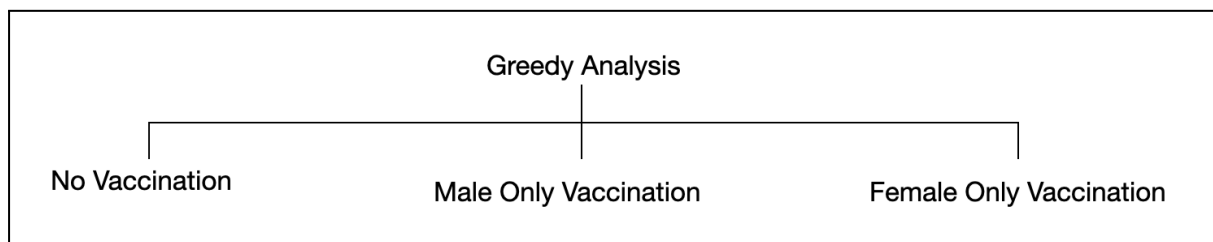
Ideas mentioned in this section are trying to analyze and explore the potential of HPVsim software, which is vital for inferring the pattern in the Cancer Incidence rate over the simulation years and analyzing the efficacies of various Interventions ranging from Screening and Treatment and Vaccination.

5.1 Cost-Effectiveness Analysis of Intervention Techniques

Delicately, we will delve into diverse intervention scenarios, harmonizing our endeavors to designate a singular strategy as the optimal choice. Throughout this exploration, I will articulate the challenges confronting this concept, ultimately determining if a distinct victor emerges or if a synergistic combination proves more favorable.

5.1.1 Greedy Analysis

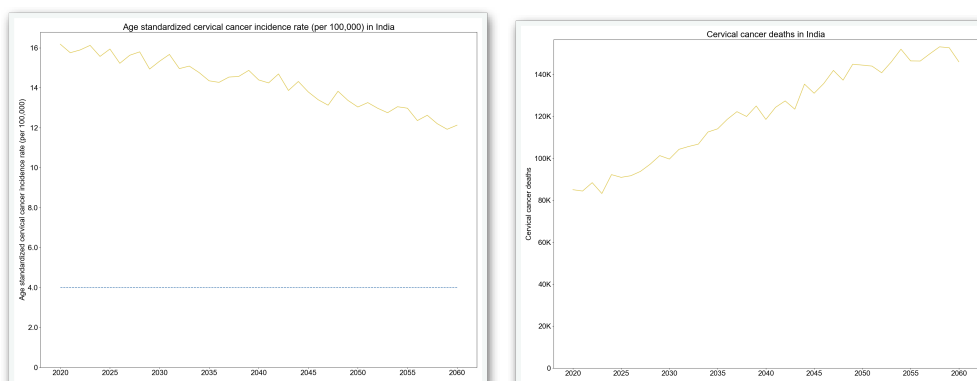
Here we will take into account various scenarios “stand-alone” to gauge their efficacies. We will then try to juxtapose interventions and their corresponding bending in the curves brought in by them.

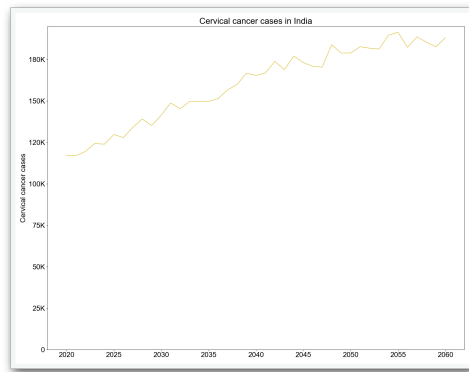


NO SCREENING

No Vaccination:

In this case, we will not perform any sort of Intervention and thus try to see what is the pattern of the Incidence rates and the death rates over time.

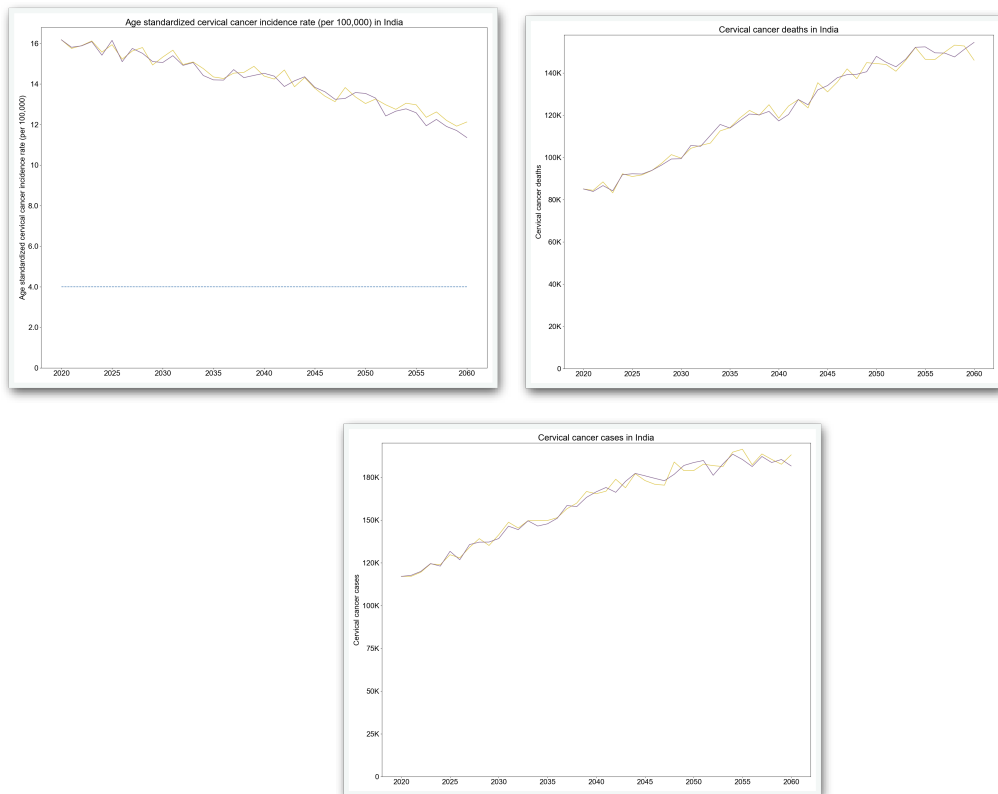




Explanation: This is the base case, where no Intervention was being carried out.

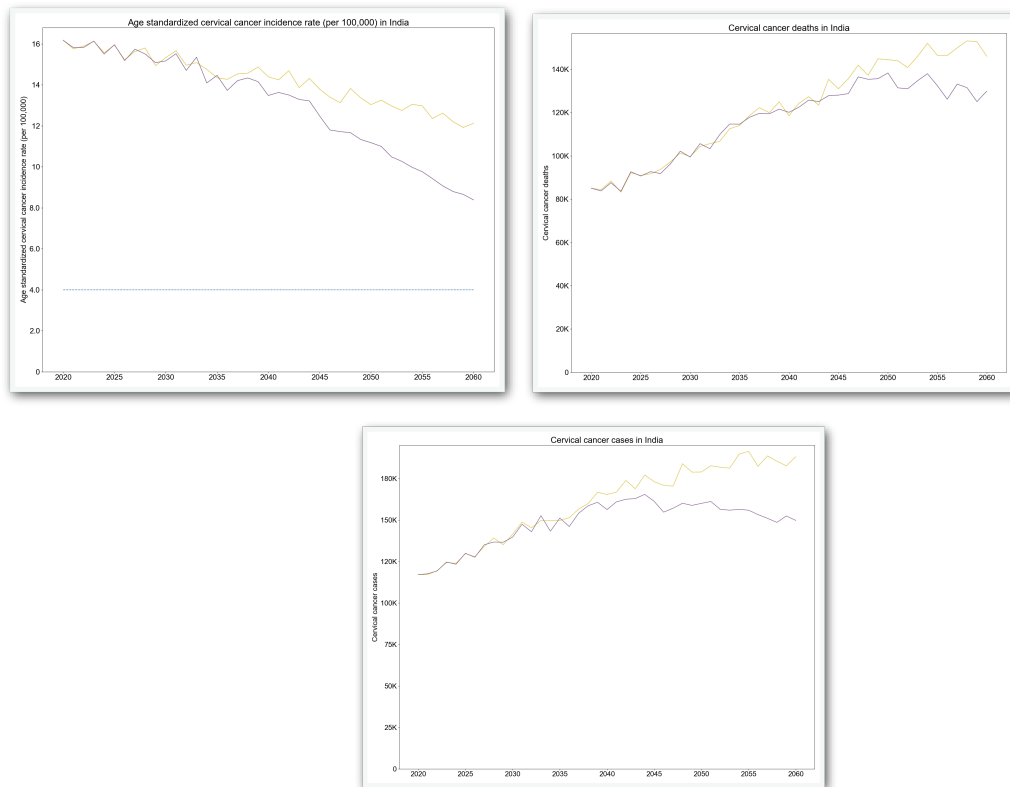
a) *Male Only Vaccination*

The objective behind carrying out such type of Intervention was to analyze whether the vaccination of males does hold any credibility on its own in bringing a decline in the new Cancer incidence cases.



b) *Female Only Vaccination*

The objective behind carrying out such type of Intervention was to analyze whether the vaccination of females holds any credibility on its own in bringing a decline in the new Cancer incidence cases.



Conclusive Remarks

1. **Vaccinating Only Males** seems to provide **No** to very **slight Improvement** only.
2. **Female Only Vaccination** provides **Drastic Improvement** in the **Incidence rates** **although** Surprisingly **Deaths** and **Cancer Cases Don't** See that Dramatic **decline**.

Inference

1. Investing under the name of something like **Gender Neutral Vaccination** should be really thought of, as **Vaccinating Males** is as good as **No Vaccination**, so to attract more Females population, blindly Male Population can't be Vaccinated, it seems.
2. Investment on **Vaccination** itself **needs** to be well **thought** off as it **doesn't** seem to be bending the **#of Deaths** and **#of Cancer Cases** Surge to **Decline**.

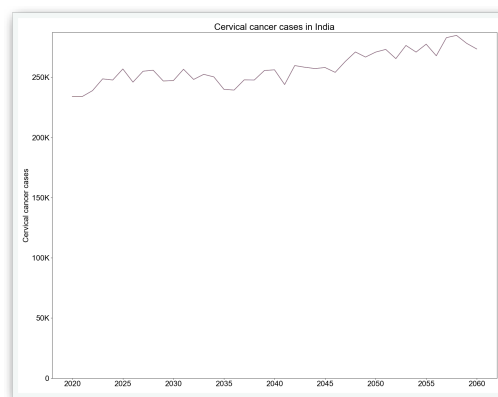
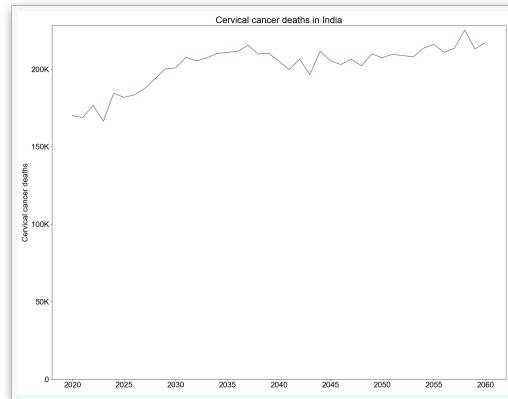
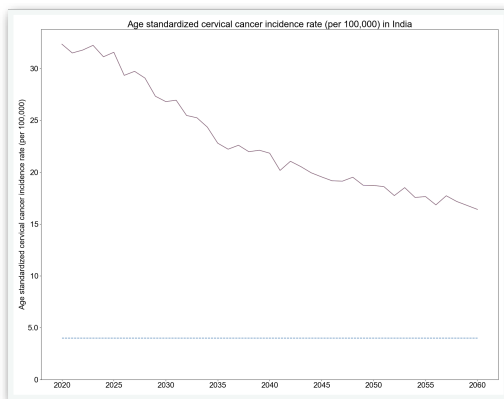
5.1.2 NON-GREEDY PRACTICE WITH SCREENING

Combining S&T with V:

NO VACCINATION ONLY SCREENING *GREEDY SETTING*

Only Screening Seems to be 'CHEAP' *in a sense than* 'VACCINATION'

```
'70% S&T cov': dict(  
    primary='hpv',  
    screen_coverage=0.7,  
)
```



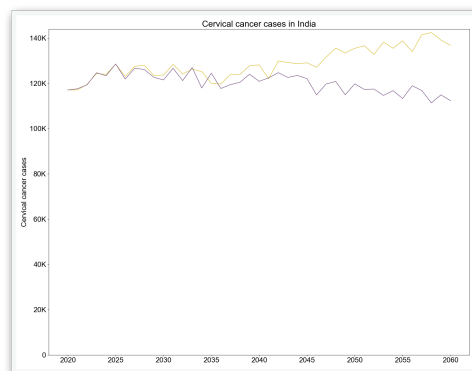
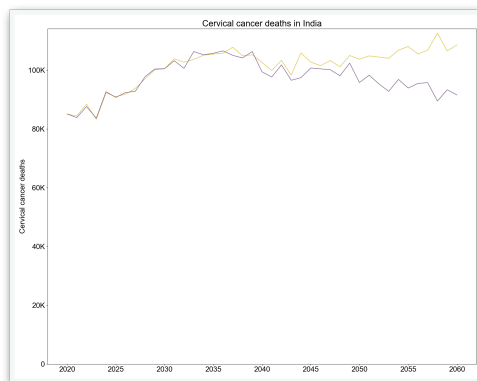
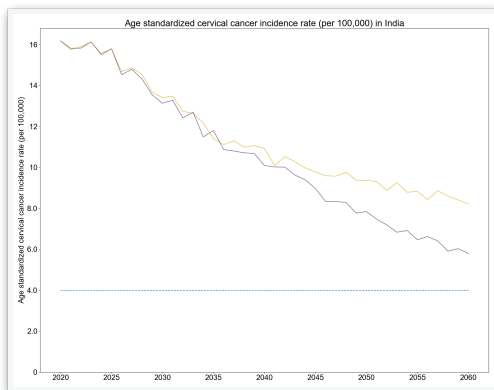
Conclusive Remarks

So 'Small' Amount of Vaccination and More Screening seems to be a Good Formula.
Regarding Gender Neutral Vaccination, Male Vaccination should be really thought of How to incorporate to motivate Females. Coz Males Vaccination doesn't seem to bend the Curve at all!

SCREENING AND VACCINATION FEMALE VACCINATION ONLY

```
'70% S&T cov': dict(  
    primary='hpv',  
    screen_coverage=0.7,  
)
```

```
'Vx, 90% cov, 9-14': dict(  
    vx_coverage=0.9,  
    age_range=(9, 14)  
)
```



COMMENT

So the only Concern seems to be Cervical Cancer deaths and Cases Doesn't seem to be decline much.

NEXT UP

1. IDENTIFY how much BENDING does GREEDY settings give which can be INCORPORATED to achieve the BENDING in NON GREEDY settings.
2. Check Whether these last 2 Graphs are Standardised or Not wrt Population Growth.
3. Check the same plotting over other COUNTRIES, whether they observe the SAME Pattern or is it only INDIA that faces this issue.

5.1.3 Formulating Optimisation Problem

A) General theme as to Why I was interested in Analysing the Male Only Vaccination?

1. I was interested in assessing how vital a role is played by S&T or Vaccination alone in bringing a bend in the Various Curves, for postulating a definitive formula of the form:

$$a \cdot (\text{Proportion of S\&T}) + b \cdot (\text{Proportion of Vaccination}) = c \cdot (\text{Proportion of bending in the Curve})$$

2. So while doing so i also tried pushing the boundaries little bit deeper in the Vaccination regime with No S&T, where i was interested in knowing how much a pivotal ratio for male as well female population should be so that it balances out the Gender Neutrality Notion, which is seen to be alleviate the Female Stigma about it while also ensuring a sound economic equilibrium since we have now extended the scope to include male vaccination as well.

So while i was in that quest, i was tinkering about such formulation in the back of my mind,

$$m \cdot (\text{Percentage Males Vaccinated}) + f \cdot (\text{Percentage Females Vaccinated}) = g \cdot (\text{Proportion of Bending in the Curve})$$

1. *Percentage of Males Vaccinated is Proportional to Investment on Male Vaccination*
2. *Percentage of Female Vaccinated is Proportional to Investment on Female Vaccination*

3. So in embarking my this pursuit, I, utilized the parameters provided during the last day of the workshop, which were tailored to India context, but the results indicated that: male-only vaccination appeared to provide little to no improvement over the scenario where no vaccination was implemented at all. This finding contrasts with data from other countries, and I find it to be rather perplexing.

Further Discussions had led to conclude:

1. The concept of a "Gender Neutral Vaccination" strategy warrants serious consideration as Vaccinating males seems to have little to no impact.
2. Another potential concern to consider delicately is the reliability of the parameters in the India Context.

So particularly as what I had learnt in those 3 days of workshop that Male Vaccination, brings in multi-faceted outcomes like, alleviating the HPV prevalence to some extent and also helps reduce the "female-only" stigma which is associated with this vaccination.

So looking at all this I was just curious, whether given a fixed amount of budget, what should be the best ratio (or rather the minimum proportion of males that needs to be vaccinated against females) for vaccination protocol, which could not only boost the Female Vaccination, which is technically the most essential part of the regime, but also give a good economic equilibria.

And by economic equilibria, I mean to say that, investment in Male Vaccination should also bring in some proportionate ROI.

And that is why I was trying to see how much bending Male-only vaccination has to offer, which then could help evaluate the efficacy of Male-Only Vaccination and thus make some conclusion regarding the "Best Ratio" for Vaccination.

Model Details

1 Optimization Model for Vaccination Protocol

In the context of vaccination strategy planning, we aim to develop a mathematical model that maximizes the Return on Investment (ROI) of Male Vaccination while considering budget constraints and maximizing female participation. The model is defined as follows:

1.1 Variables

Let:

$$\begin{aligned}x &= \text{Proportion of males to be vaccinated} \\y &= \text{Proportion of females to be vaccinated}\end{aligned}$$

1.2 Objective Function

Maximize the ROI of Male Vaccination:

$$\max \text{ROI}_{\text{Male Vaccination}} = \text{Benefit}_{\text{Male Vaccination}} - \text{Cost}_{\text{Male Vaccination}} \quad (1)$$

1.3 Constraints

1.3.1 Budget Constraint

Ensure that the total vaccination cost does not exceed the allocated budget:

$$\text{Cost}_{\text{Male Vaccination}} \cdot x + \text{Cost}_{\text{Female Vaccination}} \cdot y \leq \text{Budget} \quad (2)$$

1.3.2 Proportion Constraint

Ensure that the proportions of males and females vaccinated sum to 1 (100)

$$x + y = 1 \quad (3)$$

1.3.3 HPV Prevalence Constraint

Male Vaccination should contribute to a reduction in HPV prevalence, but not exceed a predefined limit:

$$\text{Prevalence}_{\text{Male Vaccination}} \leq \text{Prevalence}_{\text{Baseline}} \quad (4)$$

1.3.4 Reduction of Female-Only Stigma Constraint

Male Vaccination should aim to reduce the ‘female-only’ stigma associated with this vaccination, subject to a predefined threshold:

$$\text{Stigma}_{\text{Female Vaccination}} \leq \text{Stigma}_{\text{Baseline}} \quad (5)$$

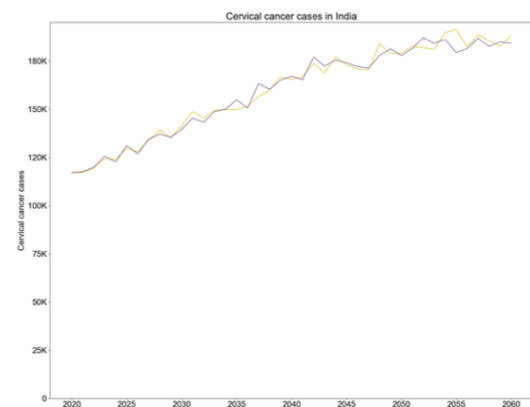
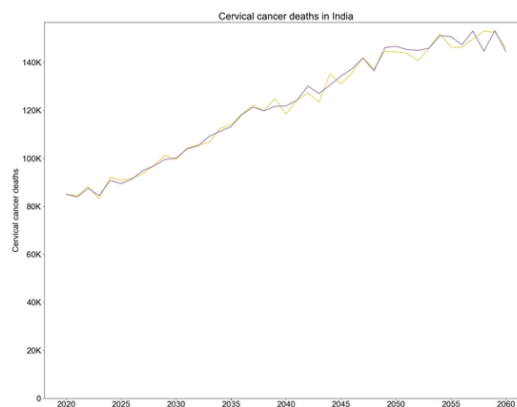
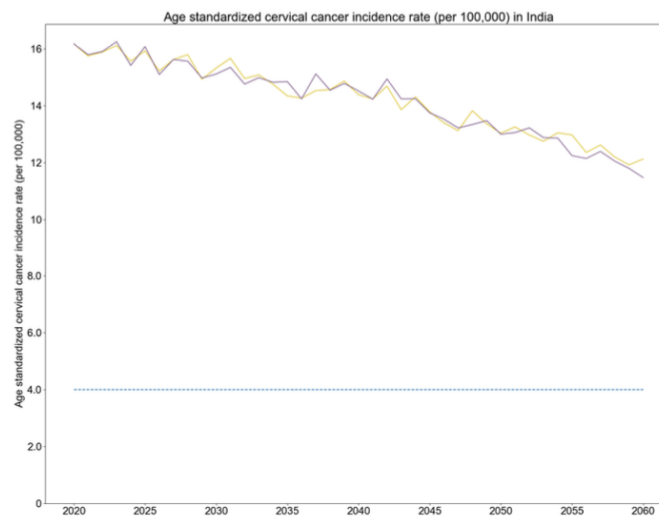
This mathematical model allows for the optimization of vaccination protocols by determining the appropriate proportion of males and females to be vaccinated while considering budget constraints, HPV prevalence reduction, and stigma reduction. Solving this model will provide insights into achieving economic equilibrium in vaccination strategies.

Some Results of running on their params:

Results

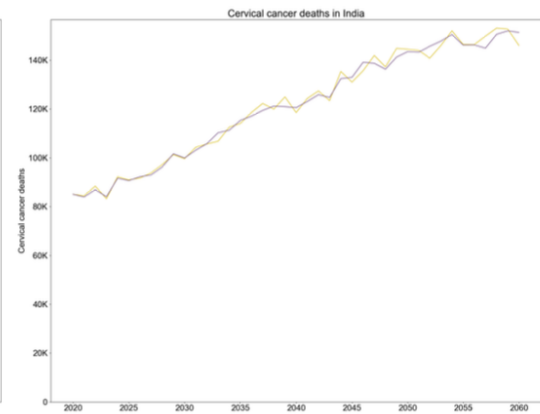
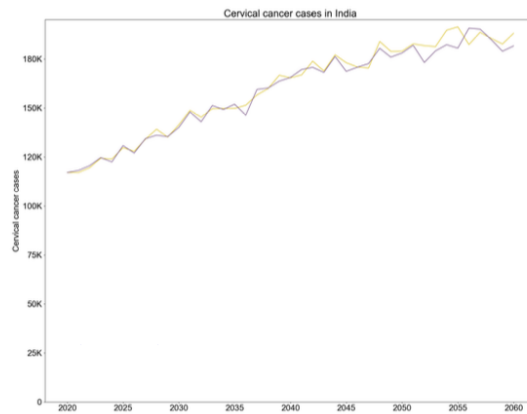
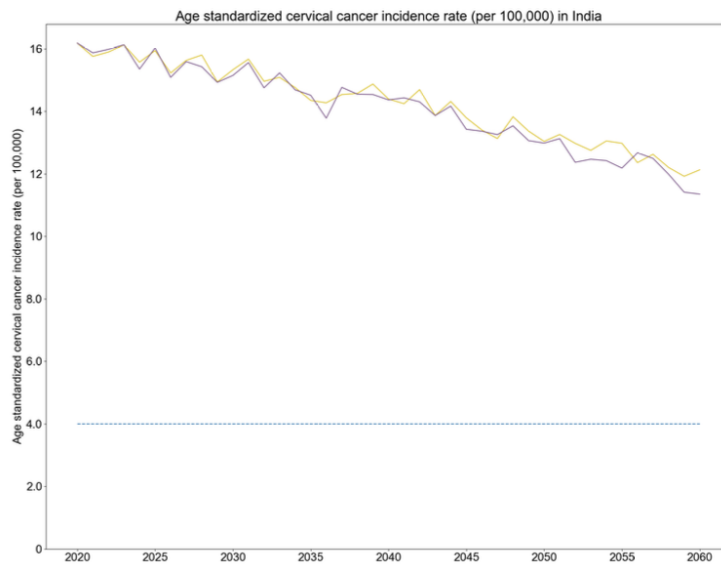
Scenario 1:

```
screen_scens = sc.objdict({  
    'No screening': dict()  
})  
vx_scens = sc.objdict({  
    'Male-Only Vx': dict(  
        vx_coverage=0.9,  
        age_range=(9, 14),  
        male_eff_redux=True # Set male_eff_redux to True for male-only vaccination  
    )  
})
```



Scenario 2:

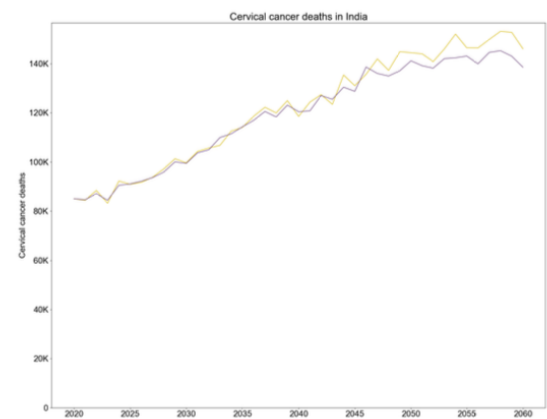
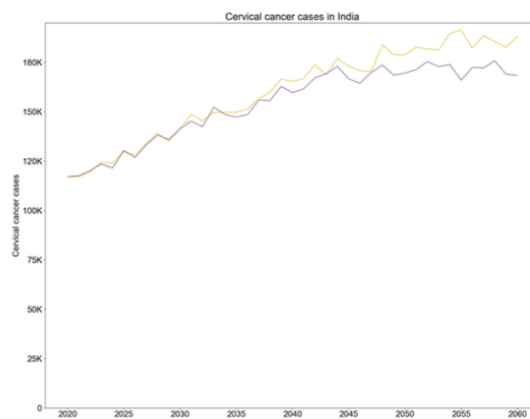
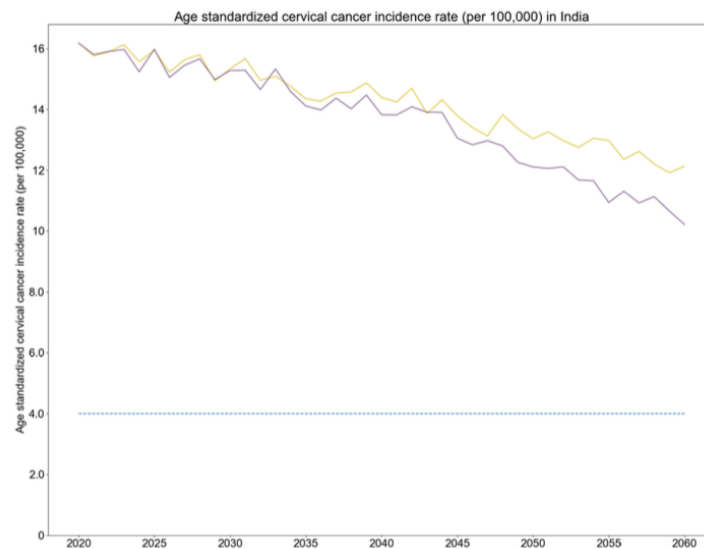
```
screen_scens = sc.objdict({
  'No screening': dict()
})
vx_scens = sc.objdict({
  'Male-Only Vx': dict(
    vx_coverage=0.97,
    age_range=(9, 14),
    male_eff_redux=True # Set male_eff_redux to True for male-only vaccination
  )
})
```



```

Scenario 3:
screen_scens = sc.objdict({
  'No screening': dict()
})
vx_scens = sc.objdict({
  'Male-Only Vx': dict(
    vx_coverage=0.90,
    age_range=(9, 22),
    male_eff_redux=True # Set male_eff_redux to True for male-only vaccination
  )
})

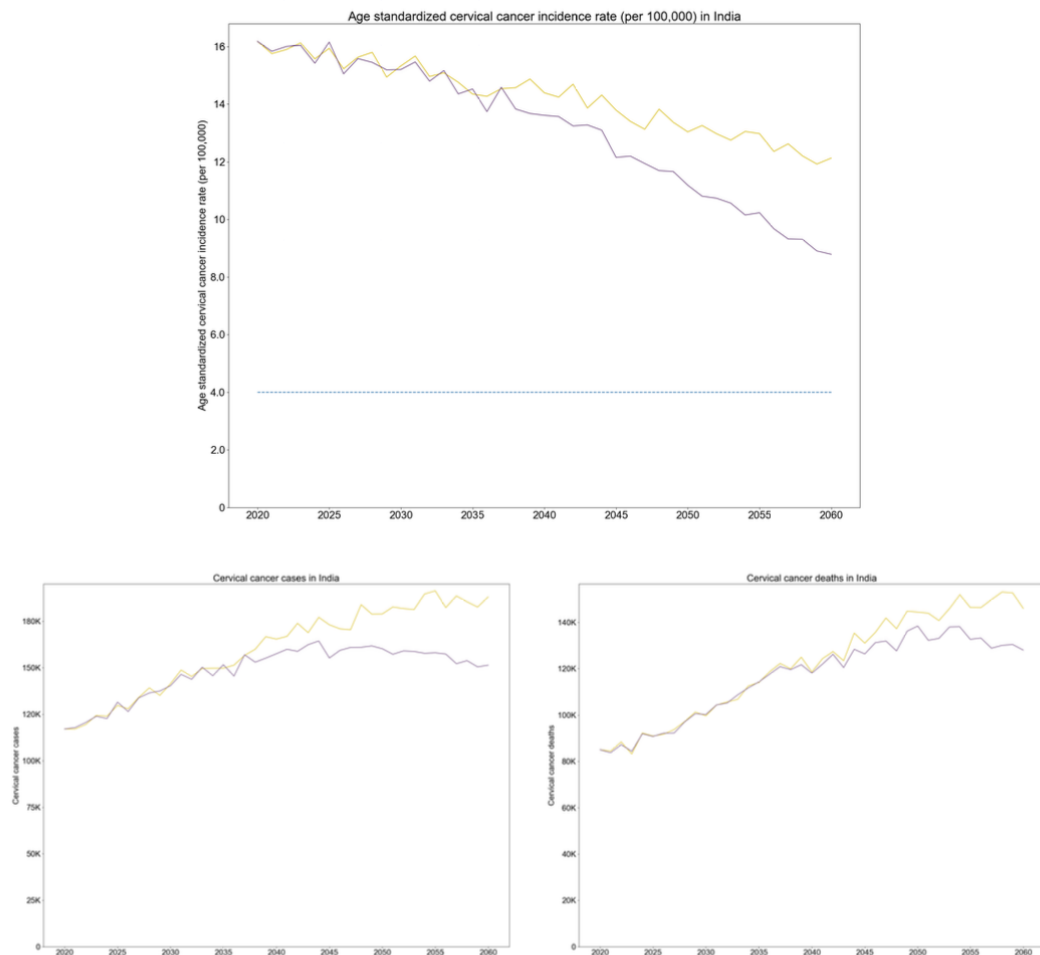
```



```

Scenario 4:
screen_scens = sc.objdict({
    'No screening': dict()
})
vx_scens = sc.objdict({
    'Male-Only Vx': dict(
        vx_coverage=0.97,
        age_range=(9, 30),
        male_eff_redux=True # Set male_eff_redux to True for male-only vaccination
    )
})

```



B) Utilizing Params found in our research.

The only parameters among what we have and what we could 'possibly' tweak, are

1. Sexual Debut

HPVsim

Males: 17.0

Female: 14.8

****They claim that NO DIRECT DATA is available.**

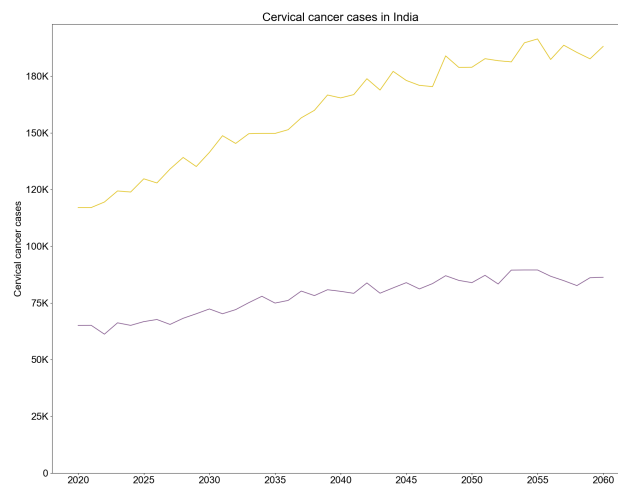
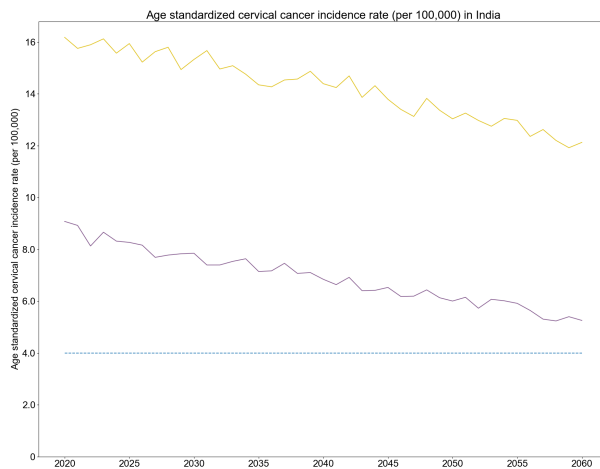
Ours

Females: 17.8 (**Need to check!**)

2. Condom Usage 5.6%

They have also taken exactly the same.

Due to the Sexual Debut Param different I **'Does'**, see some differences in plots:



** ABOVE PLOTS GENERATED FOR THIS SETTING

```
screen_scens = sc.objdict({  
  'No screening': dict()  
})
```

```

vx_scens = sc.objdict({
  'Male-Only Vx': dict(
    vx_coverage=0.9,
    age_range=(9, 14),
    male_eff_redux=True # Set male_eff_redux to True for male-only vaccination
  )
})

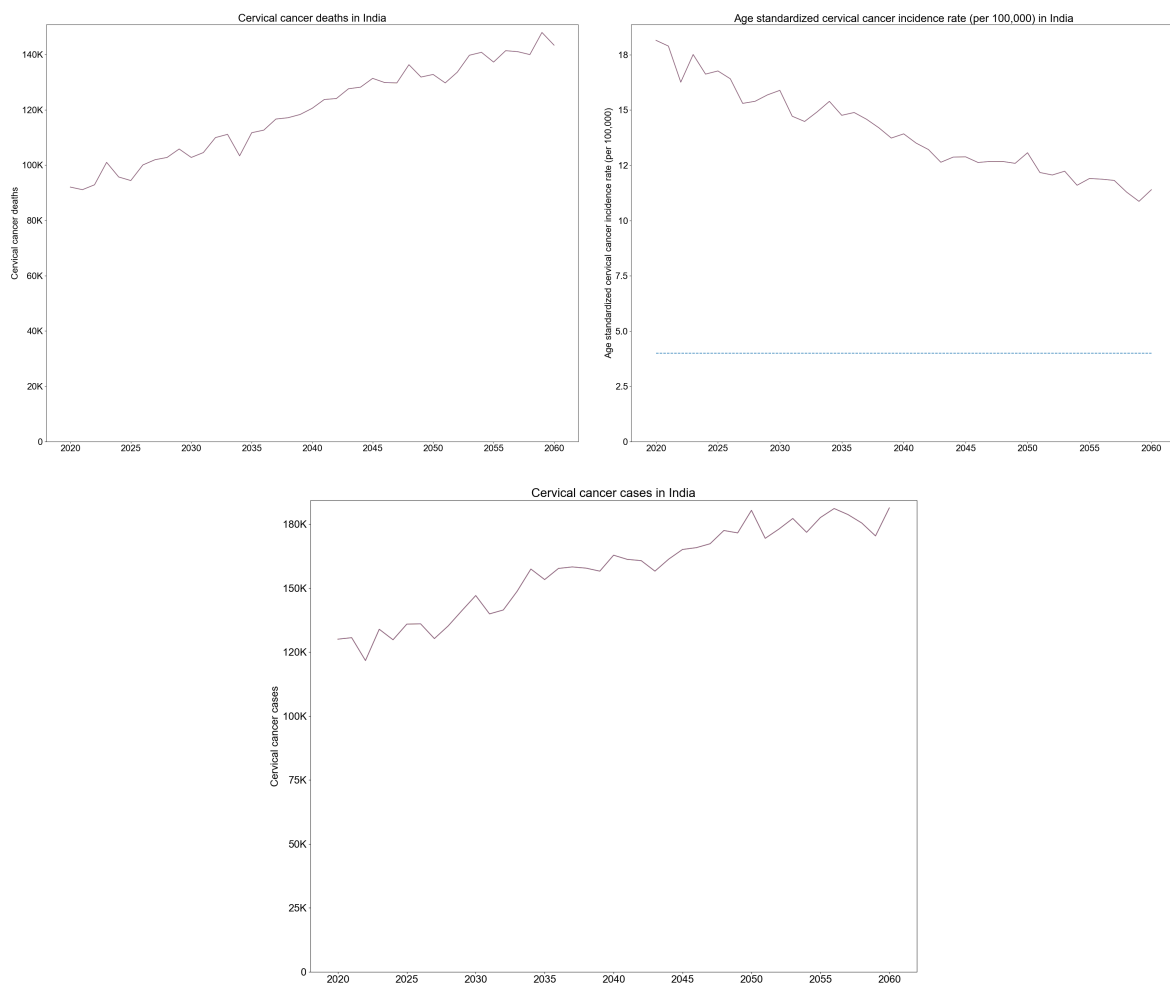
```

C) The Last 2 graphs don't appear to be standardized per population in that year, directly.

5.1.4 Results with Updated Params

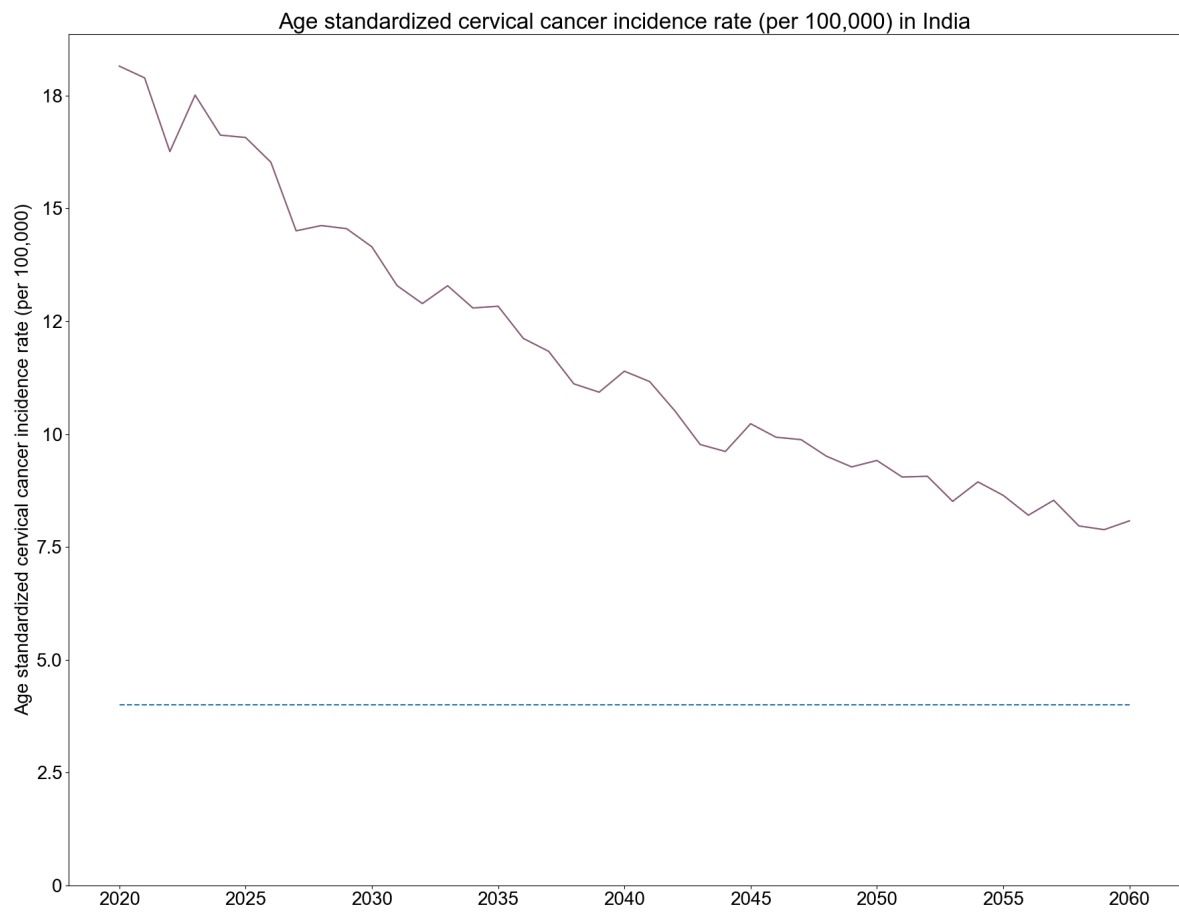
No S&T and No V

Case of NO SCREENING AND NO VACCINATION



Implementing First Algorithm of S&T

Algorithm 1

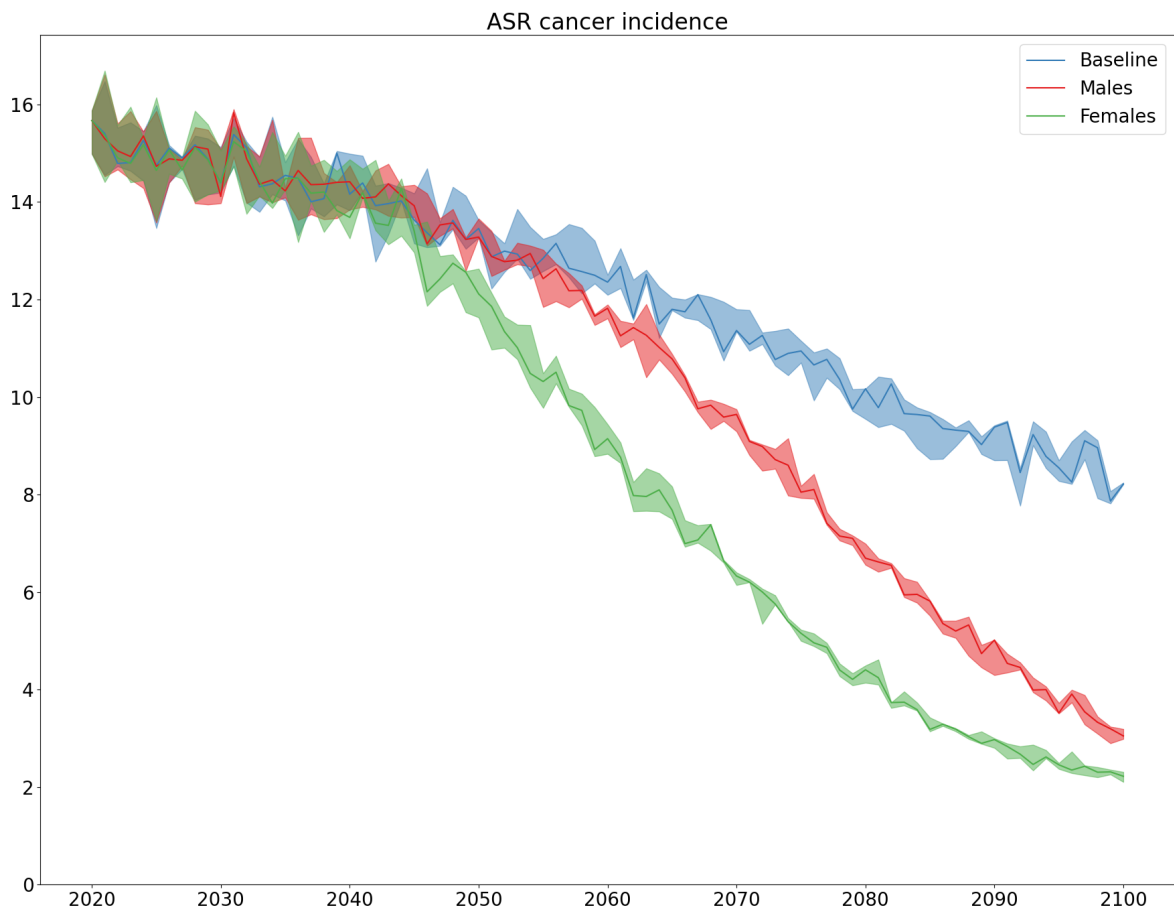


5.1.5 Challenges Faced while Implementation

Task 1

Male-Only Vaccinating Works!

Vaccinating Males vs Females And No Screening



Task 2

It was observed that except for Algorithm 1, No Algorithm is Working.

5.1.6 Issues Resolved.

There was a missing component, which was identified in the aftermath of the failed implementation of 6 out of 7 algorithms, as mentioned in the paper.

Issue Faced: *All the algorithms are behaving the same, so I wrote a custom function for every Algorithm, this led to Algorithms 1,2 and 5th to work nicely. But this approach failed to produce output for the rest of the Algorithms wherein for algorithms 3, and 7, I found that there's no way to implement them at all!*

Concluded the above based on a typical error that was seen in the HPVsim, when I was writing custom functions for each algorithm as summarised below:

Observation from error:

1. error indicates that the key 'positive' is not found in the outcomes object, which is why a KeyError is raised.

2. In pars_scenarios.py at line 184 i.e. this line of code (given below) the key 'positive' should exist in the dictionary but this key is likely missing.

```
screen_positive = lambda sim:
```

```
sim.get_intervention('screening').outcomes['positive']
```

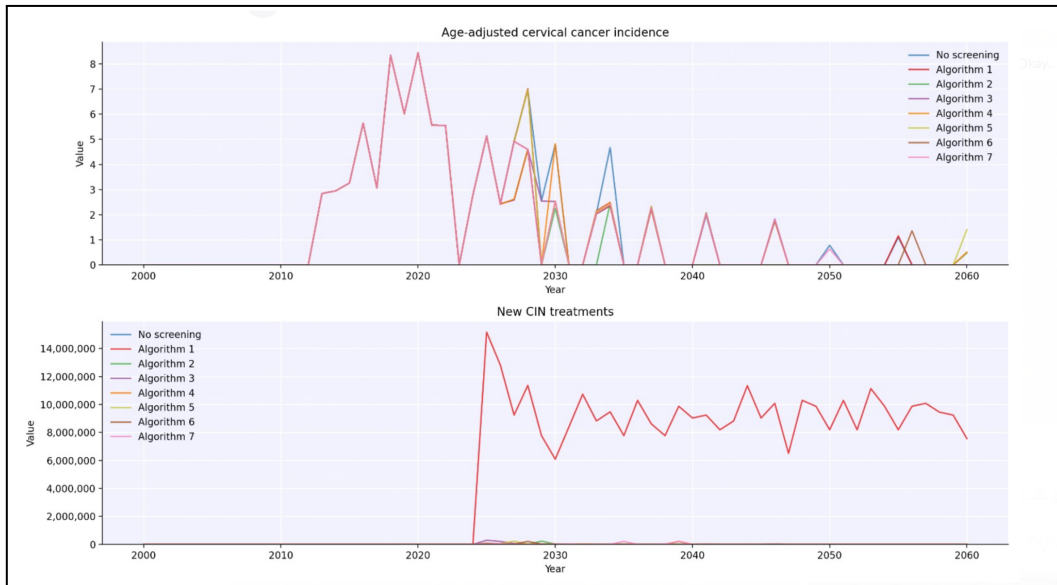
Raised this Concern with BMGF, to which they **agreed that precisely an important component (listed above) within HPVsim was not coded at all!**

So I am happy to tell that the code I wrote was indeed Right!

Now they have incorporated the **missing component** in HPVsim, and the code works!

Below is the Output for 7 Screening and Treatment, purely in the India Context,

5.1.7 S&T Algorithms in India Context



5.1.8 Formalising the Problem

To find Optimal percentages α and β
Which will lead to following Course of Action

$$\alpha \cdot (S \& T) + \beta \cdot (V) \quad \beta \ll \alpha$$

Where we will want the following to hold

1. CIR must decline by $K\%$
2. $\alpha \cdot (\text{Cost of } S \& T) + \beta \cdot (\text{Cost of } V) \leq \text{Budget } B$
3. $V = m \cdot (V_{\text{maleVacc}}) + f \cdot (V_{\text{femaleVacc}}) \quad f \gg m$

So in this

We would like to fix K

And obtain Corresponding
 α and β

Considering the above Objective

Next Course of Action

To build a Model, which will try to learn the pattern of CIR in Nations which have Good amount of work already done in this field.

Based on that we would be interested in Divergence Factor δ

$$\alpha \cdot (S \& T) + \beta \cdot (V) \implies K\% \text{ decline CIR} \quad \text{US}$$

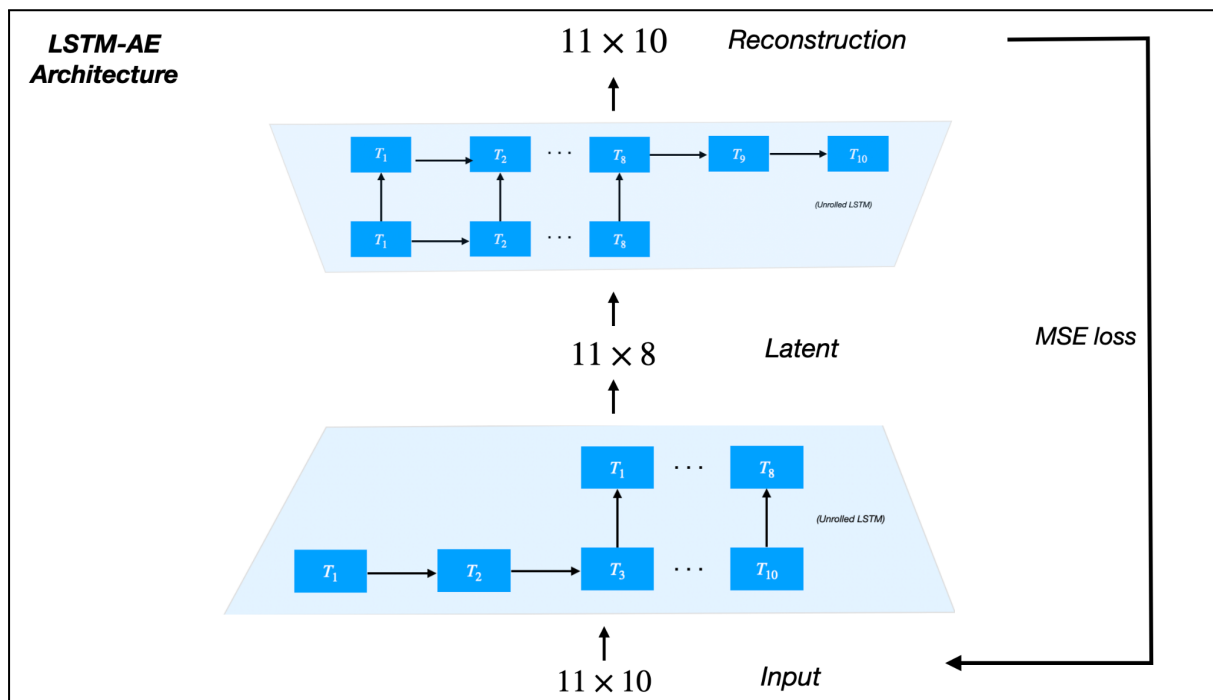
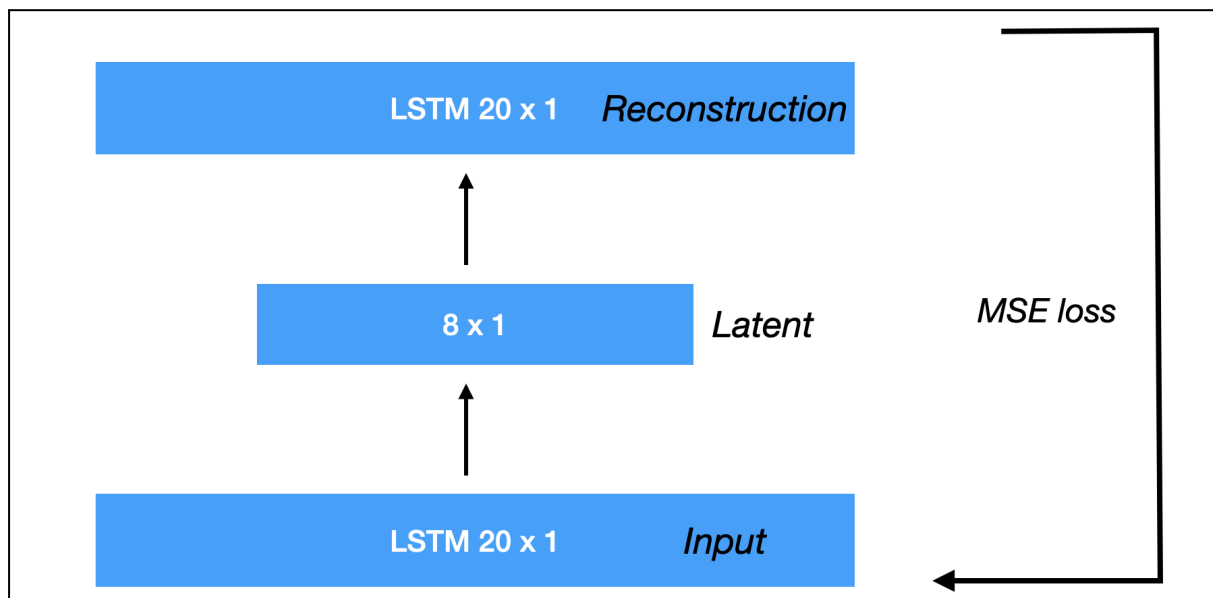
$$\alpha_{\text{delta}} = \alpha \cdot \delta \quad \beta_{\text{delta}} = \beta \cdot \delta$$

$$\alpha_{\delta} \cdot (S \& T) + \beta_{\delta} \cdot (V) \implies K\% \text{ decline CIR} \quad \text{INDIA}$$

That's how we could implement Transfer Learning in Our Context

5.1.9 LSTM-Based AutoEncoder for Anomaly Detection in Timeseries.

MSE loss over trained dataset over US, giving 0.15 loss and with another random data its giving 1.076 as loss.



6. HPVsim Projected Workflow

6.1 Comprehensive Analysis of Various Vaccination Strategies

6.1.1 Analysis of Prophylactic Vaccines

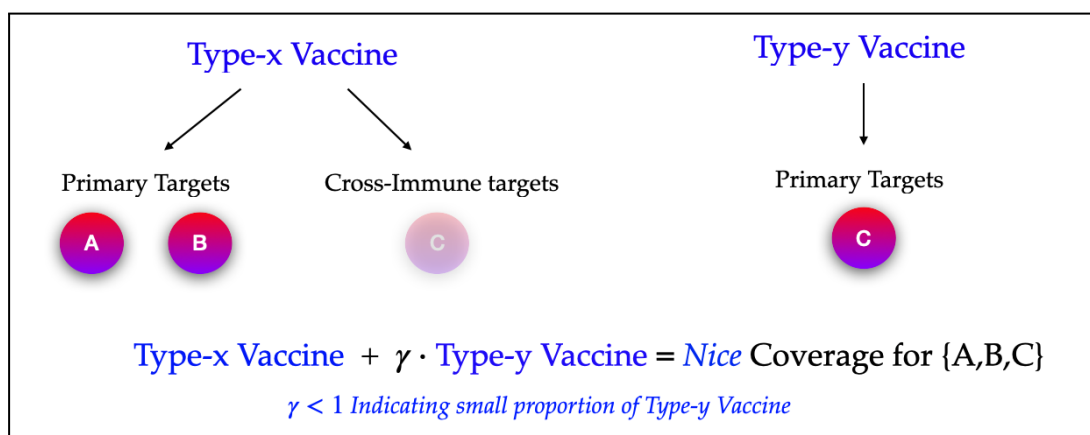
1. Cost-Effectiveness Analysis of Various Vaccines.

To compare various vaccination strategies with the amount of investment that needs to be done by adjusting the appropriate MRPs of the Vaccines compared to the overall declines in Cancer Cases which is perceived as a byproduct.

2. Exploring Cross Immunity Capabilities of Vaccines.

HPV Vaccines are identified to provide some lower level of cross-immunity towards genotypes it's not specifically targeted at.

So let's say there are 2 vaccines type-x and type-y, and say the type-x vaccine is designed to target genotype a, b , but it also provides some efficacy for genotype c , and the type-y vaccine covers genotype c only. Then a nice vaccination strategy could be formulated where More of type-x could be taken and less of type-y could be provided. The below figure shows an illustration of the Idea:



In essence, to analyse what extra could be covered just by providing some particular Vaccine, which further could be coupled with some other Vaccines with appropriate proportion.

3. Identifying the right age group and the best ratio for Vaccinating Males and Females.

Female Vaccination is the demand of the hour! But vaccinating males also helps bring a decline in cancer incidence cases, at least to some extent, thus identifying the best ratio of Male Vaccination and female vaccination becomes vital.

4. Analyse Efficacy of Routine delivery and Campaign delivery of Vaccines.

Compare various strategies by varying targeted age groups and time frames of delivery to identify what works the best!

6.1.2 Analysis of Therapeutic Vaccines

1. Juxtapose the 2 Approaches Orchestrated by WHO viz. Mass-Vaccination and Screen-and-Vaccinate.

Contrast ideas about whether to mass-vaccinate without positive test result Vs incorporating the therapeutic vaccination within some existing Screening Strategy.

2. Cost-Effectiveness Analysis of Various Therapeutic Vaccination Strategies.

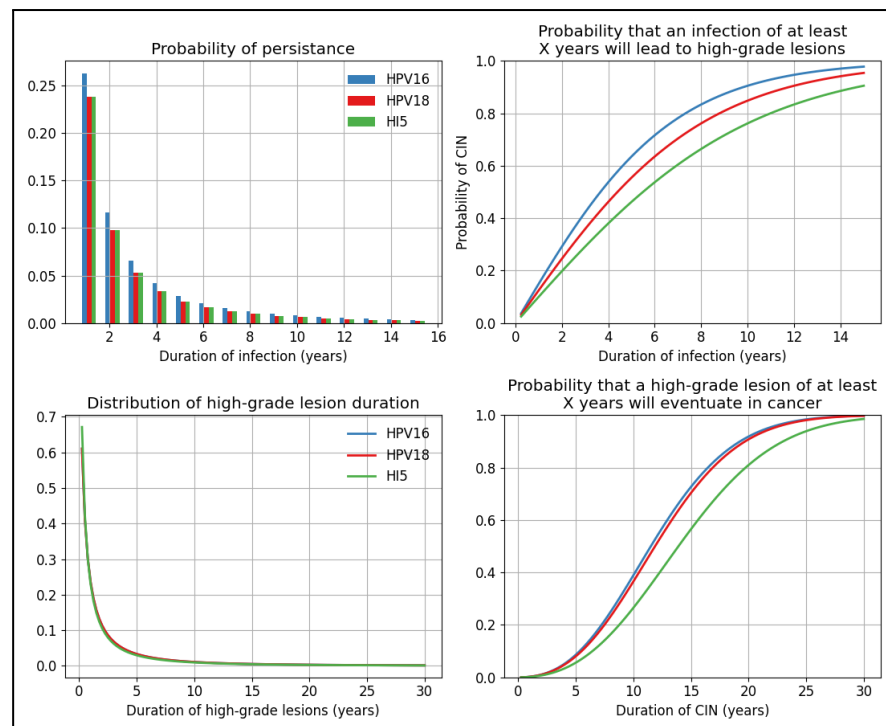
6.1.3 Contrasting Prophylactic Vaccination with Therapeutic Vaccination

1. Cost-Effective Analysis of Prophylactic vs Therapeutic Vaccination Strategies

2. Identify the trade-offs in each class of strategy which can help supplement/complement each other.

6.2 Experimental analysis of some of the Findings in Manuscript

6.2.1 To analyze the duration variability based on various Vaccine Intervention Strategies.



Figure

The plots show the duration of X quantity where X could be {CIN, infection, High Grade Lesions}.

Experiment

An experiment to see how these various durations could be varied by performing various types of interventions, which could in turn help policymakers make the right design choices to tackle this disease.

Example

As a toy example, one particular experiment could be to select the Prophylactic Vaccine in the right proportion (let's say the type-x and type-y vaccine example

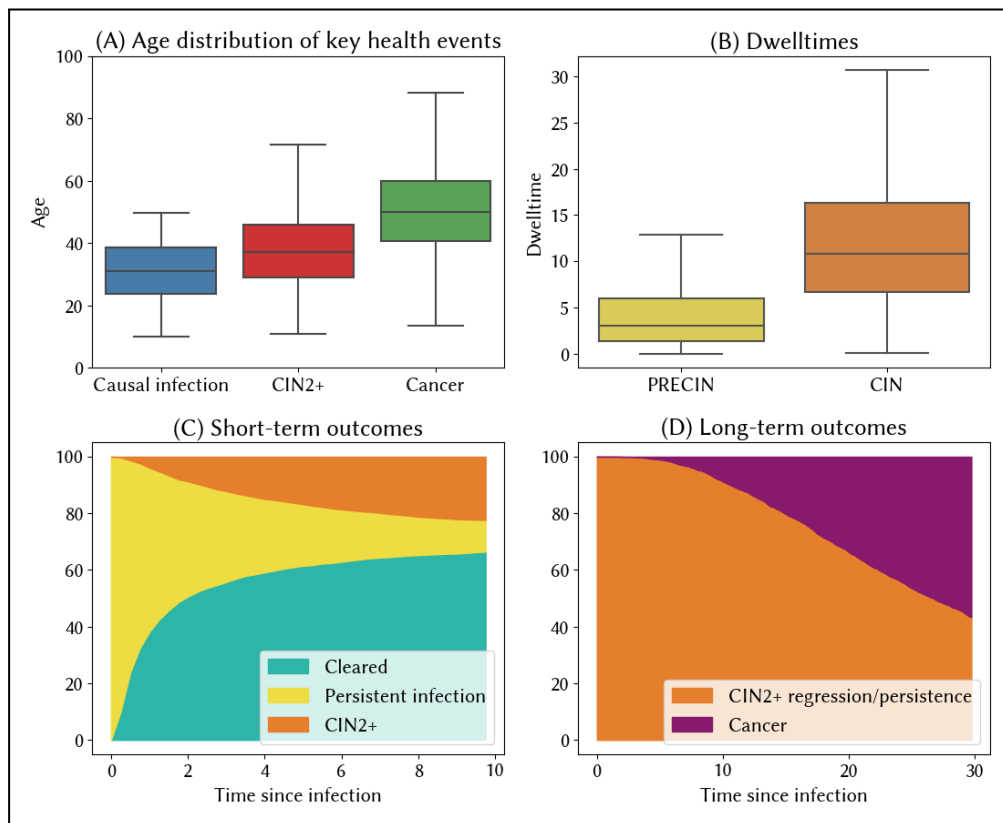
discussed above motivating the cross-immune efficacy), which could not just impact the target HPV genotype but also some other genotype to some lower extent (whose impact could be made even stronger by using another vaccine but only sparingly which targets that HPV genotype primarily).

Such an experiment could be extended by the policymaker and would give more flexibility in making the right design choice, suitable for its particular case.

Extended-Experiment

Rather than just generalizing the above plot for one geographical region as a whole, the above analysis could be made on a more granular level, i.e. targeting particularly impacted locations, where there is a surge of cases, thus making the whole set-up more versatile.

6.2.2 To analyze the various stage gaps based on various Vaccine Intervention Strategies.



Experiment

On performing various Vaccine intervention Strategies how do Age Distribution and dwell time variations change over time?

Explanation

The idea here is to see the efficacy of the various Vaccine Intervention strategies in various Geographies within India and, over time how the durations seen in the above plot change.

The right design of the Vaccine could be used to keep control over people from various geographical regions going into a cancerous(severe) state.

6.2.3 Validating Multiscale Modeling Approach in a more robust fashion.

Idea

As per the experiment details given in the Manuscript, a more rigorous explanation through more experiments needs to be seen for why the reduction of the CoV was so much. ***It would be good to investigate more on how so much reduction comes by switching to multiscale modeling.***

Questions

Some Questions that need to be thought off (and experimented):

1. Whether the number of classes in the multiscale modeling is thresholded by the total population's size (sum of weights of the agents initialized in the model), is that the reason for such reduction? If yes then how are we taking into account the growing population over the years of simulations (like basically the birth rates and death rates.)
2. How about tracking some of the level-1 agents to some extent in case they are severely indulged in disease progression? Because keeping every such level-1 agent completely devoid of transmission model seems to introduce a bias or skewness in the progression? Maybe now the CoV might go up? Seems like there is a little more potential for experimental analysis.
3. Initializing such a model seems a little tricky/difficult as how to club the population into various agent categories and then fixing the right amount for the multiscale to cover all the population and that too in a simulated

environment where the population also changes (based on birth rates and death rates) over the time span of simulation.

So an experiment that compares various initializations and analyses their efficacies would be better.

7. Case Study: ECG Brain Signal Analysis

7.1 Problem Formulation

Time Series Anomaly Detection using LSTM Autoencoders with PyTorch in Python

Data

The [dataset](#) contains 5,000 Time Series examples (obtained with ECG) with 140 timesteps. Each sequence corresponds to a single heartbeat from a single patient with congestive heart failure.

An electrocardiogram (ECG or EKG) is a test that checks how your heart is functioning by measuring the electrical activity of the heart. With each heart beat, an electrical impulse (or wave) travels through your heart. This wave causes the muscle to squeeze and pump blood from the heart. [Source](#)

We have 5 types of hearbeats (classes):

- Normal (N)
- R-on-T Premature Ventricular Contraction (R-on-T PVC)
- Premature Ventricular Contraction (PVC)
- Supra-ventricular Premature or Ectopic Beat (SP or EB)
- Unclassified Beat (UB).

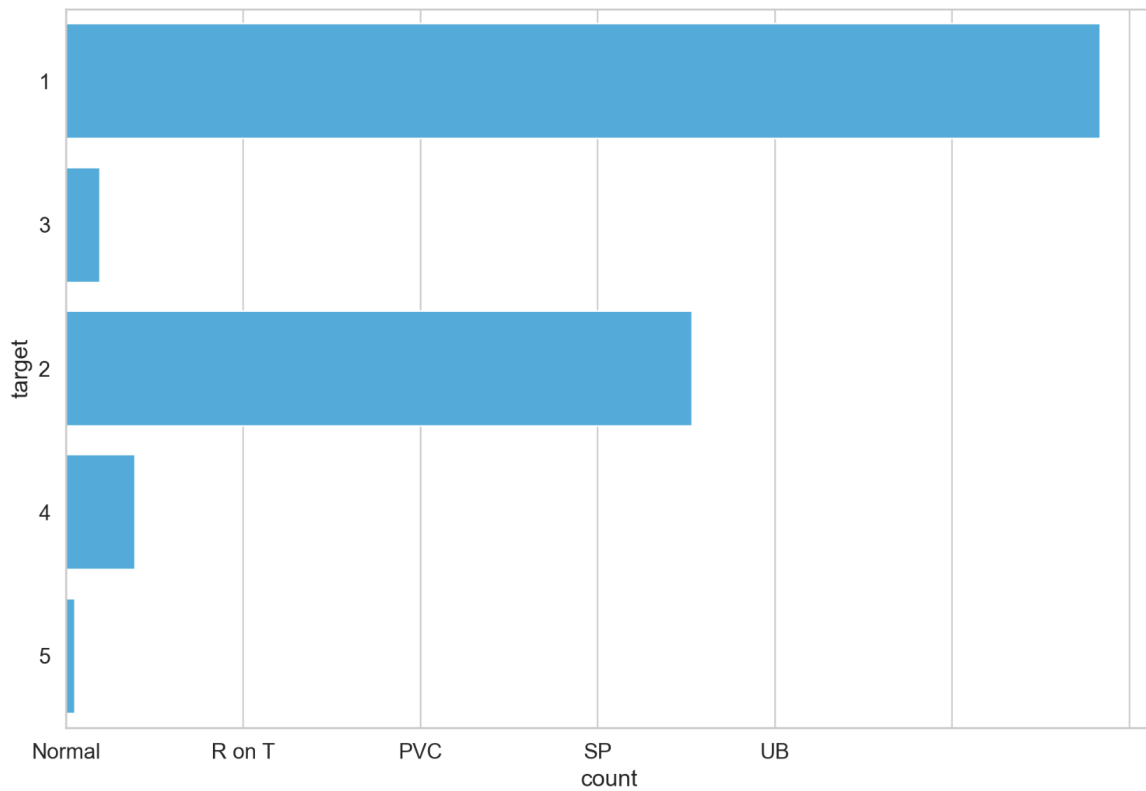
Assuming a healthy heart and a typical rate of 70 to 75 beats per minute, each cardiac cycle, or heartbeat, takes about 0.8 seconds to complete the cycle. Frequency: 60–100 per minute (Humans) Duration: 0.6–1 second (Humans) [Source](#)

The dataset is available on my Google Drive. Let's get it:

Look in to the Data set

7.2 Dataset

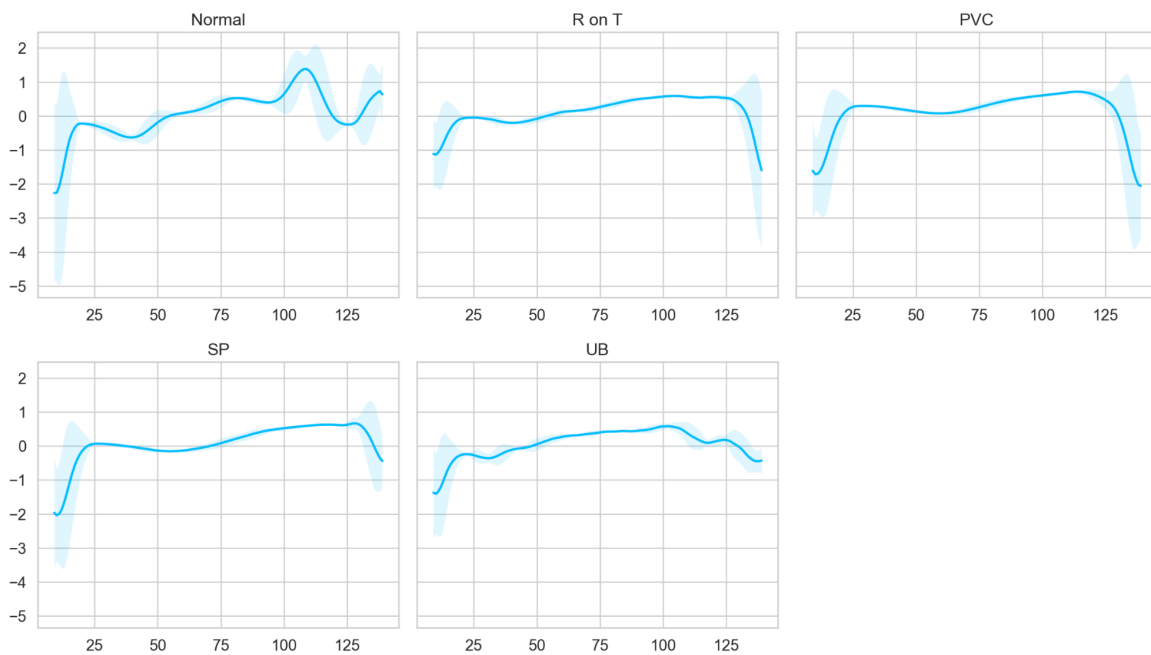
att2	att3	att4	att5	att6	att7	att8	att9	att10	...	att132	att133	att134	att135	att136	att137	att138	att139	att140	target
-1.048520	-3.394356	-4.254399	-4.162834	-3.822570	-3.003609	-1.799773	-1.500033	-1.025095	...	0.945178	1.275588	1.617218	1.580279	1.306195	1.351674	1.915517	1.672103	-1.039932	1
-3.770552	-4.267091	-4.256133	-3.515288	-2.554540	-1.699639	-1.566366	-1.038815	-0.425483	...	1.008577	1.024698	1.051141	1.015352	0.988475	1.050191	1.089509	1.465382	0.799517	1
3.365038	-3.695653	-4.094781	-3.992549	-3.425381	-2.057643	-1.277729	-1.307397	-0.623098	...	1.085007	1.467196	1.413850	1.283822	0.923126	0.759235	0.932364	1.216265	-0.824489	1
-1.671363	-3.236131	-3.966465	-4.067820	-3.551897	-2.582864	-1.804755	-1.688151	-1.025897	...	0.545222	0.649363	0.986846	1.234495	1.280039	1.215985	1.617971	2.196543	0.023843	1
-3.270123	-3.778723	-3.977574	-3.405060	-2.392634	-1.726322	-1.572748	-0.920075	-0.388731	...	0.828168	0.914338	1.063077	1.393479	1.469756	1.392281	1.144732	1.668263	1.734676	1



The normal class, has by far, the most examples. This is great because we'll use it to train our model.

Original Dataset, trend of normal and abnormal data.

7.3 Ground-Truth Trend



LSTM Autoencoder

The **Autoencoder's** job is to get some input data, pass it through the model, and obtain a reconstruction of the input. The reconstruction should match the input as much as possible. The trick is to use a small number of parameters, so your model learns a compressed representation of the data.

In a sense, Autoencoders try to learn only the most important features (compressed version) of the data. Here, we'll have a look at how to feed Time Series data to an Autoencoder. We'll use a couple of LSTM layers (hence the LSTM Autoencoder) to capture the temporal dependencies of the data.

To classify a sequence as normal or an anomaly, we'll pick a threshold above which a heartbeat is considered abnormal.

Reconstruction Loss

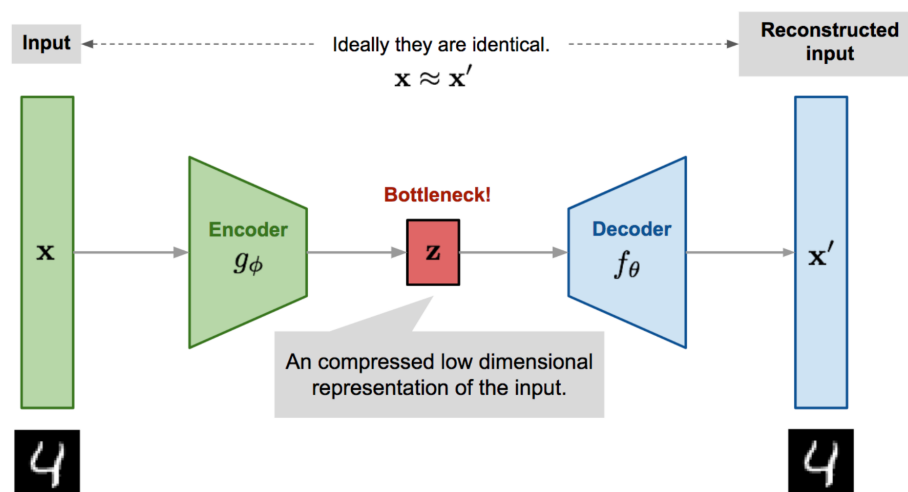
When training an Autoencoder, the objective is to reconstruct the input as best as possible. This is done by minimizing a loss function (just like in supervised learning). This function is known as *reconstruction loss*. Cross-entropy loss and Mean squared error are common examples.

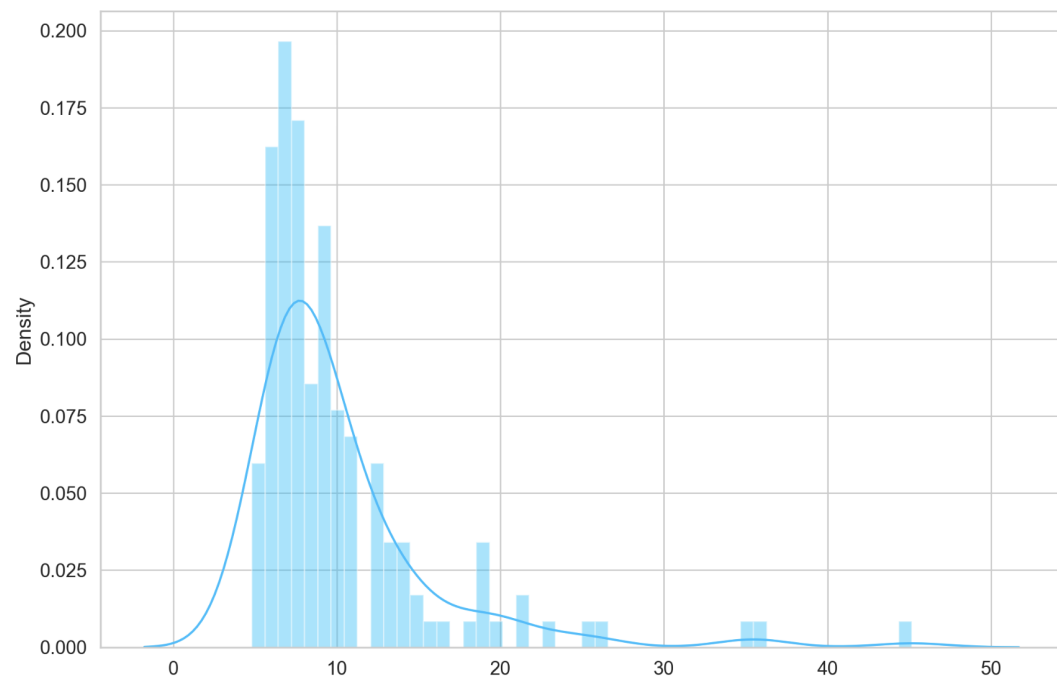
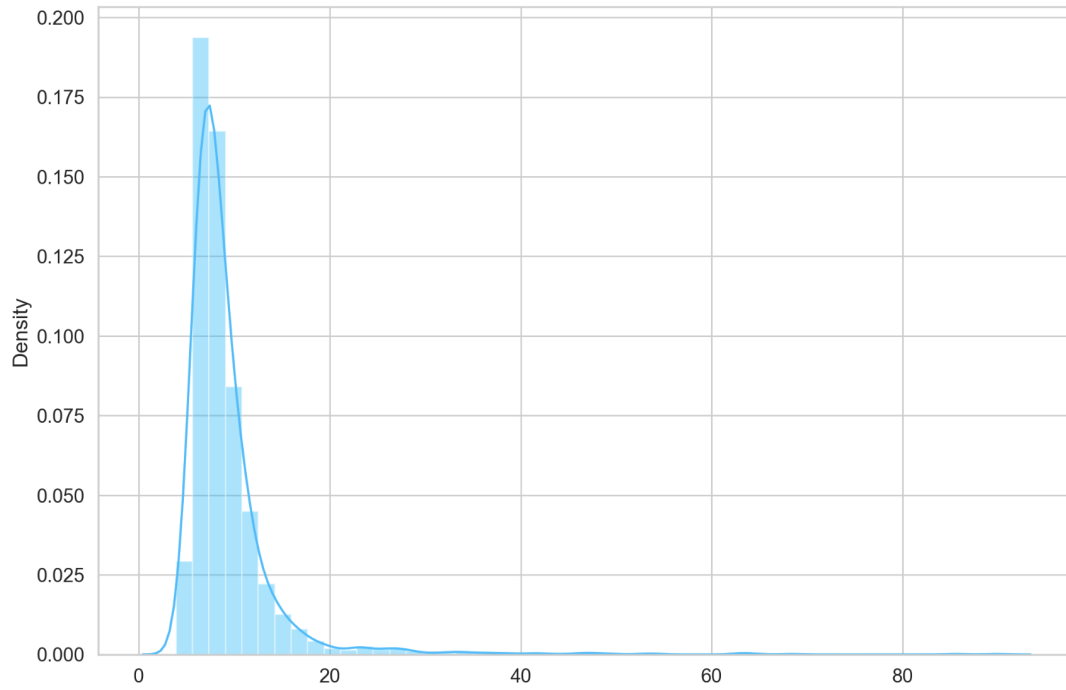
Anomaly Detection in ECG Data

We'll use normal heartbeats as training data for our model and record the *reconstruction loss*. But first, we need to prepare the data:

7.4 Model

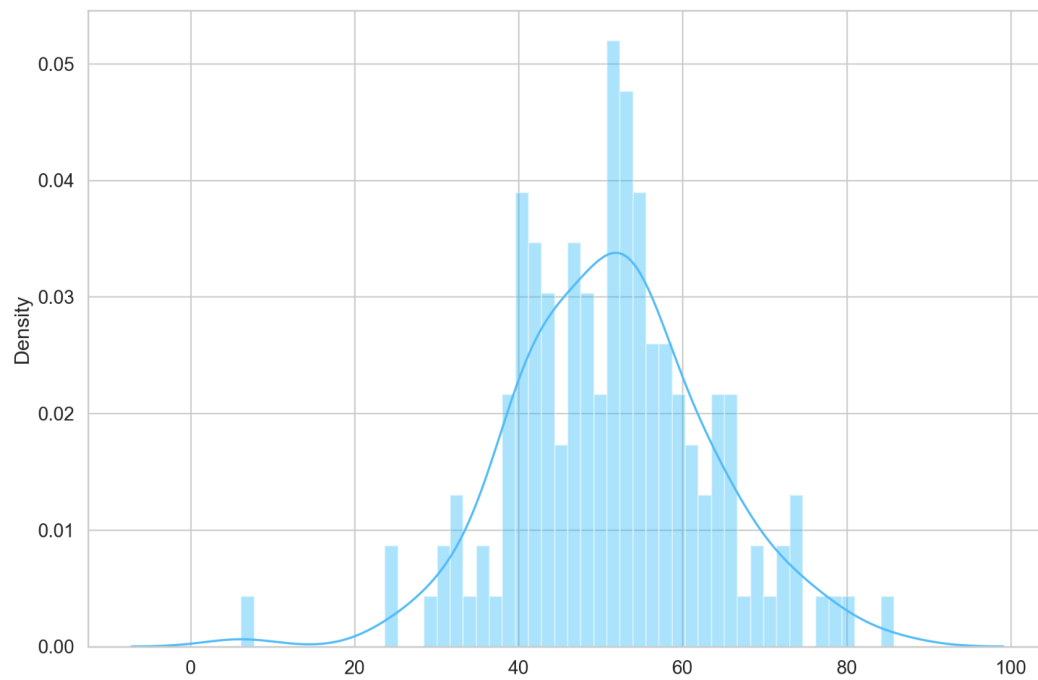
LSTM Autoencoder





Analysis to identify the right threshold:

7.5 Threshold Setting



7.6 Pattern Analysis

Observe the trend difference between what is normal and what is abnormal.

The idea is not to identify the divergence factor δ based on the reconstruction loss flagged.

